

RENANCE

FEDERAL UNIVERSITY OF TECHNOLOGY, MINNA

CHM 111 — Past Questions & Materials

by Renance

Course: CHM 111

Institution: Federal University of Technology, Minna

Papers in this pack: 1

Levels: 100

This study pack was compiled and restructured by Renance for CHM 111 students at Federal University of Technology, Minna. It brings together 1 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

Study online: renance.app

Get the app: Google Play — [dev.renance.renance](https://play.google.com/store/apps/details?id=dev.renance.renance)

Generated by Renance AI

Paper 1 of 1 — CHM111 (Acid, Base and Salt)

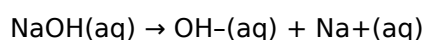
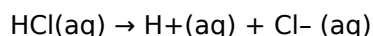
Level 100 • First Semester • Handout

Acids, Bases, pH and Buffer

There are three concepts of acids and bases in current use. Each has its own peculiar advantages. The student should understand all the three concepts: (a) Arrhenius concept, (b) Bronsted-Lowry concept and (c) Lewis concept.

Arrhenius concept

Savante Arrhenius (1884) proposed his concept of acids and bases. According to this concept, an acid is a compound that releases H^+ ions in water; and a base is a compound that releases OH^- ions in water. For example, HCl is an Arrhenius acid and NaOH is an Arrhenius base.



Bronsted-Lowry concept

In 1923 J.N. Bronsted and J.M. Lowry independently proposed a broader concept of acids and bases. According to this theory, an acid is any molecule or ion that can donate a proton (H^+) a base is any molecule or ion that can accept a proton. For brevity we can say that an acid is a proton donor

while a base is a proton acceptor.

Examples of Bronsted acids and bases

(1) HCl gas and H_2O . When dry HCl gas dissolves in water, each HCl molecule donates a proton to a water molecule to produce hydronium ion

Thus HCl gas is a Bronsted acid and water that accepts a proton is a Bronsted base.

(2) HCl and Ammonia, NH_3 . HCl gas reacts with ammonia (NH_3) to form solid NH_4Cl . HCl is a proton donor and hence a Bronsted acid, while NH_3 is a proton acceptor and a Bronsted base.

(3) Calcium oxide and H_2O . When calcium oxide is dissolved in water, it is converted to calcium hydroxide. Here a water molecule donates a proton to oxide ion, O^{2-} , and is a Bronsted acid.

The oxide ion accepts a proton and gives $2OH^-$ ions, hence is a Bronsted base.

1

Conjugate Acid-Base pairs

In an acid-base reaction the acid (HA) gives up its proton (H^+) and produces a new base (A^-).

The new base that is related to the original acid is called a conjugate (meaning related) base.

Similarly the original base (B^-) after accepting a proton (H^+) gives a new acid (HB) which is called a conjugate acid. A hypothetical reaction between the acid HA and the base B^- will illustrate the above

definitions.

The acid (HA) and the conjugate base (A^-) that are related to each other by donating and accepting a single proton, are said to constitute a conjugate Acid-Base pair.

Examples of Conjugate Acid-base pairs

Let us consider the reaction between HCl(aq) and $\text{NH}_3(\text{aq})$, in this case HCl (acid) and Cl^- (conjugate base) constitute one acid-base conjugate pair. Base NH_3 and the conjugate acid (NH_4^+)

comprise the second acid-base conjugate pair. It is illustrated below

Now let us consider the reaction between acetic acid and water to form the conjugate base CH_3COO^- and the conjugate acid H_3O^+ .

2

Strength Bronsted acids and bases

The strength of a Bronsted acid depends upon its tendency to donate a proton. The strength of a Bronsted base depends on its ability to accept a proton. For example, HCl is nearly 100% ionized in

water. Its reaction with water can be depicted by the equation:

A strong acid HCl gives a weak conjugate base (Cl^-); and a strong base (H_2O) gives a weak conjugate acid (H_3O^+)

Since the reaction has proceeded almost completely to the right, it means that HCl has a strong tendency to lose a proton. Also, the base H_2O has a strong ability to accept a proton. The overall situation is that the acid and base on the left are each stronger than the conjugate acid and conjugate

base on the right. That is why the equilibrium is displaced to the right. Thus we can conclude that: A strong acid has a weak conjugate base and a strong base has a weak conjugate acid.

Amphiprotic substances

Molecules or ions that can behave both as Bronsted acid and base are called amphiprotic substances. For example, with HCl , water acts as a base in accepting a proton from the acid.

Lewis concept of acids and bases

In the early 1930s, G.N. Lewis proposed even a more general model of acids and bases. According to Lewis theory, an acid is an electron-pair acceptor and a base is an electron-pair donor.

3

Lewis pictured an acid and base as sharing the electron pair provided by the base. This creates a covalent bond (or coordinate bond) between the Lewis acid and the Lewis base. The resulting combination is called a Complex. If the Lewis acid be denoted by A and the Lewis base by B , then the fundamental equation of the Lewis theory can be written as:

It may be noted that : (1) all cations or molecules short of an electron-pair act as Lewis acids; and (2) all anions or molecules having a lone electron-pair act as Lewis bases.

Examples of Lewis reactions

[1] Between H^+ and NH_3 : Proton (H^+) is a Lewis acid as it can accept an electron-pair. Ammonia molecule ($:\text{NH}_3$) has an electron-pair which it can donate and is a Lewis base. Thus the Lewis reaction between H^+

[2] Between H^+ and OH^- : A proton (H^+) is an electron-pair acceptor and, therefore, a Lewis acid. The OH^- is an electron-pair donor and hence a Lewis base. Thus Lewis reaction between H^+ and OH^- can be written as:

[3] Between BF_3 and NH_3 : BF_3 has six valence electrons with B atom which can accept an electron-pair and is a Lewis acid. The N atom of: NH_3 has a lone electron-pair and is a Lewis base. Lewis reaction between BF_3 and NH_3 may be written as:

4

[4] Hydration of Al^{3+} . The hydration of a metal ion such as Al^{3+} is also a Lewis reaction.

Water can act both as an acid and a base

Water is an amphoteric substance. It can behave either as an acid or a base. One molecule of water transfers a proton to another molecule. There results a hydronium ion (H_3O^+) and a hydroxyl ion (OH^-).

In this reaction one molecule of water acts as a Bronsted acid and the other as a Bronsted base. The above reaction in which water molecules interact to produce a hydronium ion and a hydroxyl ion is called auto-ionization of water. It may be written as

That auto-ionization actually occurs was proved by Friedrich Kohlrausch (1840-1910). He found that even the purest water conducts electricity to a very small extent which was due to the generation of H_3O^+ and OH^- ions in water by ionization.

Water auto-ionization is most fundamental to our study of acids and bases. Obviously, H^+ ions are associated with water to give hydronium ions (H_3O^+). But for simplicity we shall generally write the dissociation equilibrium of water as

5

Relative strength of acids

The strength of an acid depends on its ability to transfer its proton (H^+) to a base to form its conjugate base. When a monoprotic acid (HA) dissolves in water, it transfers its proton to water (a Bronsted base) to form hydronium ion (H_3O^+) and a conjugate base.

For simplifying our discussion, we take

Thus we can write the equilibrium reaction as

This equation represents the dissociation of the acid HA into H^+ ion and A^- ion. Applying the Law of Mass action to the acid dissociation equilibrium, we can write

K_a is called the acid dissociation constant. In dilute solution of the acid (HA) we note that the concentration of liquid water remains essentially constant. Therefore, the terms included in the equilibrium expression. The strength of an acid is defined as the concentration of H^+ ions in its aqueous solution at a given temperature. From the equilibrium, it is evident that the concentration of H^+ ions, $[H^+]$, depends on the value of K_a . Therefore, the value of K_a for a particular acid is a measure of its acid strength or acidity.

In any aqueous solution of a strong acid, practically all the original acid (HA) is dissociated and the value of K_a is large. On the other hand, a weak acid in aqueous solution is dissociated to a

very small extent and the value of K_a is also small. Thus in general we can say that the value of acid dissociation constant is large for a strong acid while it is small for a weak acid.

6

Relative strength of bases: According to the Arrhenius concept, a base is a substance which produces OH^- ions in aqueous solution. The basic properties of such a substance are due to these hydroxyl ions. Let us consider a base BOH whose dissociation can be represented as

Applying the Law of Mass action to the above equilibrium we can write the equilibrium expression as

K_b is called the base dissociation constant or base ionization constant. The strength of a base is defined as the concentration of OH^- ions in its aqueous solution at a given temperature.

7

From the equilibrium expression, it is evident that the concentration of OH^- ions, $[\text{OH}^-]$, depends on the value of K_b . Therefore, the value of K_b for a certain base is a measure of its base strength. In the aqueous solution of a strong base, practically all the original base is dissociated and the value of K_b is large. In the case of a weak base, it is dissociated in aqueous solution to a very small extent and the value of K_b is also small.

The pH of solutions

A knowledge of the concentration of hydrogen ions (more specifically hydronium ions) is of the greatest importance in chemistry. Hydrogen ion concentrations are typically quite small numbers.

Therefore, chemists report the hydrogen ion concentration of a solution in terms of pH. It is defined as the negative of the base-10 logarithm ($\log$) of the H^+ concentration. Mathematically it may be expressed as $\text{pH} = -\log [\text{H}^+]$ where $[\text{H}^+]$ is the concentration of hydrogen ions in moles per litre. Alternative and more useful forms of pH definition are:

The pH concept is very convenient for expressing hydrogen ion concentration.

It was introduced by Sorensen in 1909. It is now used as a general way of expressing other quantities

also, for example,

8

pH Scale

In order to express the hydrogen ion concentration or acidity of a solution, a pH scale was evolved. The pH is defined as

The hydrogen ion concentration of different acidic solutions were determined experimentally. These were converted to pH values using the above relations. Then these pH values were computed

on a scale taking water as the reference substance. The scale on which pH values are computed is called the pH scale.

Water dissociates to H^+ and OH^- ions to a very small degree so that we have the equilibrium.

We can write the equilibrium expression as

Since water is so little dissociated, the concentration of undissociated molecules, $[H_2O]$, is presumed to be constant. Therefore from expression (2) we can write

K_w is called the water dissociation constant or the water ionisation constant.

When the concentrations of H^+ and OH^- ions in water are expressed in mole per litre, the value of K_w

found experimentally is 1.0×10^{-14} . From the equation it is obvious that one molecule of water dissociates to give one H^+ ion and one OH^- ion. This means that the concentration of H^+ and OH^- ions in pure water is equal. Using the expression, we have

Thus the H^+ ion and OH^- ion concentrations in pure water are both $10^{-7} \text{ mol l}^{-1}$ at 25°C and it is said to be neutral. In acidic solution, however, the concentration of H^+ ions must be greater than $10^{-7} \text{ mol l}^{-1}$. Similarly in a basic solution, the concentration of OH^- ions must be greater than $10^{-7} \text{ mol l}^{-1}$. Thus we can state:

9

Expressing the $[H^+]$ in terms of pH for the different solutions cited above, we get what we call the pH scale. On this scale the values range from 0 to 14. Since pH is defined as $-\log [H^+]$ and the hydrogen ion concentration of water is 10^{-7} , the pH of water is 7. All solutions having pH less than 7 are acidic and those with pH greater than 7 are basic.

As shown by the pH scale, pH decreases with the increase of $[H^+]$. The lower the pH, higher is the $[H^+]$ or acidity.

In any aqueous solution, the product of $[H^+]$ and $[OH^-]$ is always equal to K_w . This is so irrespective of the solute and relative concentrations of H^+ and OH^- ions. However, the value of K_w depends on temperature. At 25°C it is 1.0×10^{-14} . Thus,

Each of $[H^+]$ and $[OH^-]$ in pure water at 25°C is 10^{-7} . The concentration of H^+ and OH^- ions are expressed in gram moles per litre. The concentration of H^+ and OH^- ions can be calculated from the expressions:

10

The measurement of pH

The pH of a given solution can be measured with the help of an apparatus called pH meter.

This consists of a voltameter connected to two electrodes. (a) A standard electrode of known potential; and (b) A special electrode (the probe) enclosed in a glass membrane that allows migration

of H^+ ions. The glass case contains a reference solution of dilute HCl.

A glass electrode coupled with standard calomel electrode for determining pH

The two electrodes are dipped in the solution to be tested. If this solution has a different pH from the solution in the probe, an electrical potential results. Thus the potential between the standard

electrode and the glass electrode varies with the pH of the solution under test. This potential is recorded by an inbuilt potentiometer of the pH meter. The potentiometer reading is automatically converted electrically to a direct reading of the pH of the unknown solution. Knowing the pH of the solution its hydrogen ion concentration can be calculated.

Determination of pH of a solution using glass electrode

A half-cell is set up with the test solution as electrolyte. The emf of the cell depends on the concentration of H^+ ions or pH of the solution. The emf of the half-cell is determined by coupling it with another standard half-cell and measuring the emf of the complete cell. The commonly used standard electrode is the glass electrode.

A glass electrode is immersed in the solution of unknown pH. It is coupled with a standard calomel electrode (SCE) as shown in Figure. The emf of the complete cell can be determined experimentally.

11

A glass electrode coupled with standard

Glass electrode Calomel electrode

calomel electrode for determining pH

Calculations: The potential of the glass electrode, E_G , at $25^\circ C$ is given by equation

The value of the potential of calomel electrode is known while E_{cell} can be found experimentally. Therefore, we can find pH of a given solution if E°_G is known. It can be determined by using a solution of known pH in the cell and measuring E_{cell} . This value of E°_G is constant for a particular glass electrode and can be used for any subsequent determinations of pH of unknown solutions with the help of equation.

The potential of the cell, E_{cell} , cannot be measured using ordinary potentiometer or voltmeter as the resistance of the glass membrane is very high and the current small. Therefore, an electronic voltmeter is required which reads pH directly.

12

Merits and demerits of Glass electrode: A glass electrode is universally used because

[1] It is simple to operate. [2] It is not easily poisoned. [3] Its activity is not affected by strong oxidising and reducing agents. [4] Since E°_G depends on a particular glass electrode used, it is not a

universal constant and also changes with time. Hence a glass electrode only compares pH values while the hydrogen electrode measures pH absolutely.

What is a Buffer Solution?

It is often necessary to maintain a certain pH of a solution in laboratory and industrial processes. This is achieved with the help of buffer solutions, buffer systems or simply buffers. A buffer solution is one which maintains its pH fairly constant even upon the addition of small amounts of acid or base.

In other words, a buffer solution resists (or buffers) a change in its pH. That is, we can add a small amount of an acid or base to a buffer solution and the pH will change very little. Two common types of buffer solutions are:

[1] A weak acid together with a salt of the same acid with a strong base. These are called Acid buffers e.g., $CH_3COOH + CH_3COONa$.

[2] A weak base and its salt with a strong acid. These are called Basic buffers.

e.g., $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$.

Let us illustrate buffer action by taking example of a common buffer system consisting of solution of acetic acid and sodium acetate ($\text{CH}_3\text{COOH}/\text{CH}_3\text{COONa}$).

since the salt is completely ionized, it provides the common ions CH_3COO^- in excess.

The common ion effect suppresses the ionisation of acetic acid. This reduces the concentration of H^+ ions which means that pH of the solution is raised. The buffer solution maintains fairly constant pH and the changes in pH could be described as marginal.

How a Buffer Operates?

Now we proceed to discuss how the addition of a small amount of HCl or NaOH to the buffer solution affects its pH. The pH of the buffer is governed by the equilibrium

The buffer solution has a large excess of CH_3COO^- ions produced by complete ionisation of sodium acetate,

13

[1] Addition of HCl: Upon the addition of HCl, the increase of H^+ ions is counteracted by association with the excess of acetate ions to form unionized CH_3COOH . Thus the added H^+ ions are

neutralised and the pH of the buffer solution remains virtually unchanged. However owing to the increased concentration of CH_3COOH , the equilibrium shifts slightly to the right to increase H^+ ions.

This explains the marginal increase of pH of the buffer solution on addition of HCl.

Mechanism of Buffer action of an acid buffer

[2] Addition of NaOH. When NaOH is added to the buffer solution, the additional OH^- ions combine with H^+ ions of the buffer to form water molecules. As a result the equilibrium shifts to the right to produce more and more H^+ ions till practically all the excess OH^- ions are neutralised and the

original buffer pH restored. However, a new equilibrium system is set up in which $[\text{CH}_3\text{COOH}]$ is lower than it was in the original buffer. Consequently $[\text{H}^+]$ is also slightly less and pH slightly higher than the buffer pH values.

Operation of a Basic buffer as $\text{NH}_4\text{OH}/\text{NH}_4\text{Cl}$ can also be explained on the same lines as of an acid buffer upon addition of HCl the H^+ ions combine with OH^- ions of the buffer to form water molecules. The equilibrium,

is shifted to the right till all the additional H^+ ions are neutralised and the original buffer pH restored.

When NaOH is added to the buffer solution, OH^- ions associate with excess of NH_4^+ ions to form unassociated NH_4OH . Thus the pH of the buffer is maintained approximately constant.

Mechanism of Buffer action of a basic buffer.

14

Calculation of the pH of Buffer Solutions

The pH of an acid buffer can be calculated from the dissociation constant, K_a , of the weak acid

and the concentrations of the acid and the salt used. The dissociation expression of the weak acid, HA, may be represented as

The weak acid is only slightly dissociated and its dissociation is further depressed by the addition of the salt (Na^+A^-) which provides A^- ions (Common ion effect). As a result the equilibrium concentration of the unionized acid is nearly equal to the initial concentration of the acid.

The equilibrium concentration $[\text{A}^-]$ is presumed to be equal to the initial concentration of the salt added since it is completely dissociated. Thus we can write the equation as

where $[\text{acid}]$ is the initial concentration of the added acid and $[\text{salt}]$ that of the salt used.

Taking negative logs of both sides of the above equation, we have

This relationship is called the Henderson equation. In a similar way, the Henderson equation for a basic buffer (pOH) can be derived.

Significance of the Henderson equation: With its help: [1] the pH of a buffer solution can be calculated from the initial concentrations of the weak acid and the salt provided K_a is given. However,

the Henderson equation for a basic buffer will give pOH and its pH can be calculated as $(14 - \text{pOH})$.

15

[2] The dissociation constant of a weak acid (or weak base) can be determined by measuring the pH of a buffer solution containing equimolar concentrations of the acid (or base) and the salt.

The measured pH, therefore, gives the value of $\text{p}K_a$ of the weak acid. Likewise we can find the $\text{p}K_b$ of a weak base by determining the pH of equimolar basic buffer.

[3] A buffer solution of desired pH can be prepared by adjusting the concentrations of the salt and the acid added for the buffer. It is noteworthy that buffer solution are most effective when the concentrations of the weak acid (or weak base) and the salt are about equal. This means that pH is close to the value of $\text{p}K_a$ of the acid (or $\text{p}K_b$ of the base).

Problems

[1] The hydrogen ion concentration of a fruit juice is 3.3×10^{-2} M. What is the pH of the juice? Is it acidic or basic?

[2] If a solution has a pH of 7.41, determine its H^+ concentration.

16

[3] Calculate the pH of 0.001 M HCl

[4] The pH of a solution of HCl is 2. Find out the amount of acid present in a litre of the solution.

[5] Find out the pH of a 0.002 M acetic acid solution if it is 2.3% ionized at this dilution.

17

[6] Find the pH of a buffer solution containing 0.20 mole per litre CH_3COONa and 0.15 mole per litre CH_3COOH . K_a for acetic acid is 1.8×10^{-5}

[7] Estimate the pH at 25°C containing 0.10 M sodium acetate and 0.03 M acetic acid $\text{p}K_a$ for $\text{CH}_3\text{COOH} = 4.57$.

[8] Calculate the pH of a buffer solution that is 0.250 M in formic acid, HCOOH , and 0.100 M in sodium formate, HCOONa . K_a for formic acid is 1.8×10^{-4} .

18

Syllabus

Acids and basis – Lowry Bronsted and Lewis concepts - Concept of pH – pH measurement using glass electrode – Dissociation constants – Buffer solution – Henderson equation for calculation of pH

Previous University Questions

[1] Outline the Lowry Bronsted and Lewis concepts of acids and bases {5 marks}

[2] Outline the measurement of pH using glass electrode {5 marks}

[3] The potential of a hydrogen gas electrode in a solution of an acid of unknown strength is 0.29 V at 298 K as measured against normal hydrogen electrode. Calculate the pH of acid solution {5 marks}

[4] The H^+ ion concentration of fruit juice is 3.3×10^{-2} M. What is the pH of the juice?

Is it acidic or basic {2 marks}

[5] 40 ml of 0.10 M NaOH solution is added to 100 ml of 0.10 M solution of acetic acid. Calculate the pH of the solution (K_a for acetic acid = 1.8×10^{-5}) {5 marks}

[6] Write a short note on buffer solution. Express Henderson equation for calculation of pH {5 marks}

ietchemistry@gmail.com

19

FEDERAL UNIVERSITY OF TECHNOLOGY MINNA

SCHOOL OF PHYSICAL SCIENCES

DEPARTMENT OF CHEMISTRY

CHM 111 GENERAL CHEMISTRY I. 2021/2022 SESSION

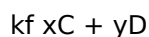
PART 1 (Released on 12/10/2021)

1.0 CHEMICAL EQUILIBRIUM.

Most chemical reactions do not go to completion. That is, when reactants are mixed in stoichiometric quantities, they are not completely converted to products. Reactions that do not go to completion or that can occur in either direction (depending on the prevailing conditions of temperature and pressure) are called reversible reactions.

Reactions takes place due to the collision of the reactant particles and immediately, products are formed. As soon as some product molecules are formed, the reversible process, that is, the formation of the reactant molecules begins to take place. At this point, equilibrium is said to be reached because the reactants and products concentrations stop changing as the forward and reverse reactions become equal. In other words, Chemical equilibrium exists when two opposing reactions occur simultaneously at the same rate. This state of equilibrium is dynamic, that is, the reaction is proceeding normally, although ordinarily it may seem to have stopped.

Consider the reaction;



Where k_f and k_r are the rate constants for the forward and reverse reactions respectively.

As the concentrations of C and D builds up, A and B begins to re-form. If the reaction begins with only species C and D, the rate of the backward reaction decreases with time and the rate of the forward reaction increases with time until the two rates are equal.

1

From the rate law;

$$\text{Rate of forward reaction} = k_f [A]^a [B]^b$$

$$a \quad b$$

$$k_r [C]^x [D]^y$$

$$x \quad y$$

Rate of reverse reaction =

$$\text{At equilibrium, the two rates are equal. Thus; } k_f [A]^a [B]^b = k_r [C]^x [D]^y$$

$$a \quad b \quad x \quad y$$

$$k_f [C]^x [D]^y$$

$$x \quad y$$

$$= = K_c$$

$$k_r [A]^a [B]^b$$

Where K_c is the equilibrium constant. The subscript 'C' indicates that the equilibrium constant is expressed in concentration units.

Equilibrium constant K is equal to a particular ratio of equilibrium product and reactant terms at a particular temperature. Although the concentration of the reactants and products are constant in a system at equilibrium, this does not mean that the system contains 50% reactants and 50% products. It is the opposing rates that are equal at equilibrium, not necessarily the concentrations. Generally, K is large for reactions that reach equilibrium with high concentration of products and small for reactions that reach equilibrium with low concentration of products.

For gaseous reactions, the partial pressures of the reacting species are used. Thus;

$$k_f P_C^x \cdot P_D^y$$

$$= = K_p$$

$$k_r P_A^a \cdot P_B^b$$

Where P_a , P_b , P_x and P_y are the partial pressures of the species A, B, C and D respectively.

1.1 The mass action expression and equilibrium constant

The particular ratio of terms written for a given reaction is called the mass action expression, Q , (or reaction quotient). The law of mass action states that at a given temperature, a chemical system reaches a state in which a particular ratio of reactant and products concentration terms has a constant value.

The reaction quotient for the general reaction above can be expressed as

$$Q =$$

$$\frac{[C]^x [D]^y}{[A]^a [B]^b}$$

$$x y$$

, where Q is the reaction quotient.

$$\frac{[A]^a [B]^b}{[C]^x [D]^y}$$

The reaction quotient has the same form as the equilibrium constant, but it involves specific values that are not necessarily equilibrium concentrations. If they are equilibrium concentrations, then $Q = K_c$. The concept of the reaction quotient is very useful. It can be used to compare the magnitude of Q with that of K_c for a reaction under given condition to decide whether the forward or the reverse reaction must occur to a greater extent to establish equilibrium.

The reaction quotient, Q, can be thought of as a measure of the progress of the reaction. When the mixture contains only reactants, the concentrations in the numerator are zero, so $Q = 0$. As the reaction proceeds to the right, the product concentrations (numerator) increase and the reactant concentrations (denominator) decrease, so Q increases to an infinitely large value when

2

all reactants have been consumed and only products remain. The value of K_c is a particular value of Q that represents equilibrium mixtures for the reaction.

If at any time $Q < K_c$, the forward reaction must occur to a greater extent than the reverse reaction for equilibrium to be established. This is because when $Q < K_c$, the numerator of Q is too small and the denominator is too large. To increase the numerator and to reduce the denominator, A and B must react to produce C and D. Conversely, if $Q > K_c$, the reverse reaction must occur to a greater extent than the forward reaction for equilibrium to be reached. When the value of Q reaches the value of K_c , the system is at equilibrium, so no further net reaction occurs. Therefore when;

$Q < K_c$, forward reaction predominates until equilibrium is established.

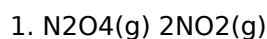
$Q = K_c$, system is at equilibrium.

$Q > K_c$, reverse reaction predominates until equilibrium is established.

1.2 Equilibrium constant expressions.

Equilibrium conditions in which all the reacting species are in the same phase are known as homogeneous equilibrium.

Examples;



In terms of pressure, K_p is given as;

2

$$K_p = \frac{P_{NO_2}^2}{P_{N_2O_4}}$$

$$K_p = \frac{P_{NO_2}^2}{P_{N_2O_4}}$$

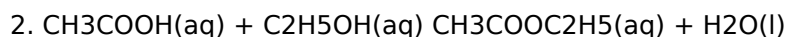
$$K_p = \frac{P_{NO_2}^2}{P_{N_2O_4}}$$

In terms of concentration, K_c is given as;

$$K_c = \frac{[NO_2]^2}{[N_2O_4]}$$

$$K_c = \frac{[NO_2]^2}{[N_2O_4]}$$

$\square \text{N}_2 \text{O}_4 \square$



$K_c =$

$\square \text{CH}_3\text{COOC}_2\text{H}_5 \square$

$\square \text{CH}_3\text{COOH} \square \square \text{C}_2\text{H}_5\text{OH} \square$

Here, water is not used in the K_c expression because the concentration of water is constant.

Note:

K_p and K_c can be used to express equilibrium constants for gases depending on whether the concentrations are expressed in terms of their partial pressures or in mol dm^{-3} .

In heterogeneous equilibria, the reactants and products are in different phases.

Example;



$K_c = \square \text{CO}_2 \square$ or $K_p = P_{\text{CO}_2}$

Also note that the concentration of the solid reactants or products is not used because their concentration is considered to be constant throughout the reaction.

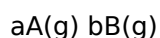
The equilibrium constant K_c , is defined as the product of the equilibrium concentrations (moles per liter) of the products, each raised to the power that corresponds to its coefficient in the balanced equation, divided by the product of the equilibrium concentrations of reactants, each raised to the power that corresponds to its coefficient in the balanced equation. Similarly, The equilibrium constant K_p , is defined as the product of the equilibrium partial pressures (in atmospheres) of the products, each raised to the power that corresponds to its coefficient in the

balanced equation, divided by the product of the equilibrium partial pressures of reactants, each raised to the power that corresponds to its coefficient in the balanced equation.

1.3 Relationship between K_c and K_p

Generally, K_c is not equal to K_p , since the partial pressures of reactants and products are not equal to their concentration expressed in mol dm^{-3} .

Consider the reaction;



where a and b are the stoichiometric coefficients. The equilibrium constant K_c is

$K_c =$

$\square \text{B} \square$

b

..... (1)

$\square \text{A} \square^a$

And

P_b

$K_p = \frac{P_b}{P_a^a}$ (2)

P_A

Where P_A and P_B are the partial pressures of species A and B.

Assuming ideal gas behaviour,

$$P_A V = n_A R T \quad \text{..... (3)}$$

$$n_A R T$$

$$P_A = \frac{n_A R T}{V} \quad \text{..... (4)}$$

$$V$$

Where V is the volume of the container in dm³

Similarly,

$$P_B V = n_B R T \quad \text{..... (5)}$$

$$n_B R T$$

$$P_B = \frac{n_B R T}{V} \quad \text{..... (6)}$$

$$V$$

Substituting equation (4) and (6) in equation (2)

We have

$$\frac{P_A}{P_B} = \frac{n_A R T}{n_B R T}$$

$$\frac{P_A}{P_B} = \frac{n_A}{n_B}$$

$$\frac{P_A}{P_B} = \frac{n_A}{n_B}$$

$$K_P = \frac{P_C^c}{P_A^a P_B^b}$$

$$= \frac{(P_A^a P_B^b)^{b-a}}{(P_A^a P_B^b)^a} \quad \text{..... (7)}$$

$$V^{b-a}$$

$$a$$

$$\frac{n_A R T}{n_A R T} = \frac{n_A R T}{n_A R T}$$

$$\frac{n_A R T}{n_A R T}$$

$$\frac{V^b}{V^a} = \frac{V^b}{V^a}$$

$$n_A$$

Now, both A and B have the units of moles per dm³ and can be replaced by [A] and [B], so

$$V^b$$

that

$$\frac{P_B^b}{P_A^a} = \frac{P_B^b}{P_A^a}$$

$$K_P = \frac{P_C^c}{P_A^a P_B^b} \quad \text{..... (8)}$$

$$\frac{P_C^c}{P_A^a P_B^b}$$

$$\frac{P_A^a}{P_A^a}$$

$$\frac{K_P}{K_C} = \frac{K_P}{K_C} (R T)^{\Delta n} \quad \text{..... (9)}$$

$$\frac{P_C^c}{P_A^a P_B^b}$$

Where $\Delta n = b - a$, that is, moles of gaseous products minus moles of gaseous reactants. Since pressure is usually expressed in atmosphere, the gas constant, R, is given by 0.0821 L.atmK⁻¹, we can write the relationship between K_p and K_c as

$$K_P = K_C (0.0821 T)^{\Delta n} \quad \text{..... (10)}$$

$$\frac{P_C^c}{P_A^a P_B^b}$$

In general, $K_P \neq K_C$, except in the special case when $\Delta n = 0$. In that case, equation (10) can be written as;

$$4$$

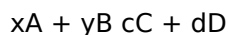
$$K_P = K_C (0.0821 T)^{\Delta n}$$

0

$$\Delta G = \Delta G^\circ + RT \ln Q \quad (11)$$

1.4 Relationship between equilibrium constant and free energy.

For any reaction such as



The free energy change, ΔG , at any temperature, T , is given by;

$$\Delta G = \Delta G^\circ + RT \ln Q$$

$$\Delta G = \Delta G^\circ + RT \ln \frac{a_C^c a_D^d}{a_A^x a_B^y}$$

$$\Delta G = 0$$

Where ΔG is the free energy of the reaction. ΔG° is the free energy of the reaction when the reactants are in their standard states. a is the activity of the reacting species. But $\Delta G = 0$ at equilibrium, therefore, the above equation becomes

$$\Delta G = 0$$

$$0 = \Delta G^\circ + RT \ln Q$$

$$\Delta G^\circ = -RT \ln Q$$

$$\Delta G^\circ = -RT \ln K$$

$$\Delta G^\circ = -RT \ln K$$

$$\Delta G^\circ = -RT \ln K$$

$$\Delta G^\circ = -RT \ln K$$

since $Q = K$, the equilibrium constant,

$$\Delta G^\circ = -RT \ln K$$

$$\Delta G^\circ = -RT \ln K$$

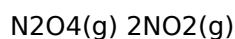
1.5 Le Chatelier's principle

This states that if one of the factors affecting the position of a system in equilibrium is changed, the system reacts in such a way as to nullify the effect of the change.

Factors affecting the equilibrium position.

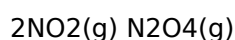
1. Temperature

Consider the homogenous equilibrium reaction



The dissociation of N_2O_4 into NO_2 gives rise to a higher number of gaseous species in the enclosed system. This means an increase in the number of moles, n , of the mixture. From the general gas equation, $PV = nRT$. n can only increase if T increases, since PV/R will be constant.

Thus, the dissociation of N_2O_4 is favoured by an endothermic reaction whereas, the reverse



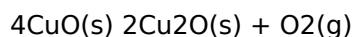
is an exothermic process.

$$K_p = \frac{P_{\text{NO}_2}^2}{P_{\text{N}_2\text{O}_4}}$$

$$K_p = \frac{P_{\text{NO}_2}^2}{P_{\text{N}_2\text{O}_4}}$$

$$K_p = \frac{P_{\text{NO}_2}^2}{P_{\text{N}_2\text{O}_4}}$$

An example of a heterogeneous equilibrium is the dissociation of copper (ii) oxide into copper (i) oxide at elevated temperature.

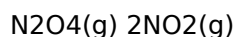


$$K_p = P_{\text{O}_2}$$

5

2. Pressure

Using the equilibrium reaction



V_1 V_2

Note that V_1 is less than V_2 . To increase V_1 to V_2 , i.e. for the forward reaction to occur, the pressure, P , of the system must be lowered.

Since

$$nRT$$

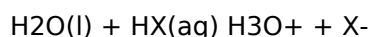
$$V =$$

$$P$$

In the above expression, the term nRT is a constant. Increasing P drives the reaction to the left, leading to the formation of more N_2O_4 .

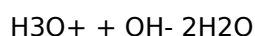
3. Concentration

Consider the dissociation of the acidic visual acid-base indicator, HX .



colour 1 colour 2

The indicator has different colours in acidic and basic media. Thus, if the concentration of H_3O^+ is increased by adding an acid to the mixture, the equilibrium will shift to the left and colour 1 becomes predominant. If on the other hand, a base (containing OH^- ions) is added to the system, the equilibrium will be in favour of the formation of the conjugate base, X^- ; the reaction will proceed in the forward direction since the reaction



removes H_3O^+ from the system, causing more HX to dissociate.

4. Catalyst

A catalyst influences both the forward and reverse reactions to the same extent and so has no effect on the equilibrium position of the reaction. It only helps the system to reach the state of equilibrium faster.

1.6 Equilibrium in acid-base media

1.6.1 The acid-base theory.

Arrhenius theory.

Acids and bases are substances that ionize in water to produce H^+ and OH^- ions respectively.

Brsted-Lowry theory;

Acids and bases are species which donate and accept protons respectively

Lewis theory classifies acids and bases as substances which can accept and donate a pair of electrons respectively.

1.6.2 Acid- base equilibria

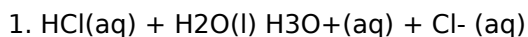
The Lowry- Brsted theory will be used in this section, since the concern here is the aqueous

systems.

According to this theory, specie having a tendency to loose a proton is an acid and will give rise to a conjugate base for having a tendency to accept a proton. Therefore, the conjugate base of a Bronsted acid is the specie that remains when one proton has been removed from the acid.

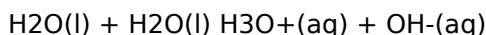
Conversely, a conjugate acid results from the addition of a proton to a Bronsted base.

6

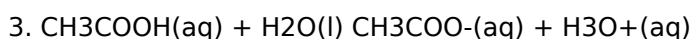


acid 1 base 2 acid 2 base 1

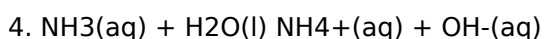
2. Self ionization of water.



acid 1 base 2 acid 2 base 1



acid 1 base 2 base 1 acid 2



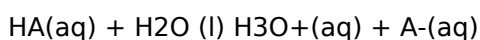
base 1 acid 2 acid 1 base 2

1.6.3 Strength of acids and bases

Strong acids and bases are completely dissociated in water while weak ones are weakly or partially ionised.

1.6.4 Dissociation constant

Consider a typical acid, HA,



Applying the equilibrium law, we obtain

$$K_a =$$

$\frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{HA}]}$, where K_a is the dissociation constant of the acid.



Water is weakly dissociated, so that essentially, the concentration is constant. We can therefore write; $K_w = [\text{H}^+][\text{OH}^-]$



For a typical base, B;

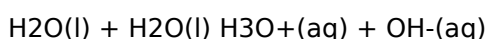
$$K_b =$$

$\frac{[\text{BH}^+][\text{OH}^-]}{[\text{B}]}$, K is the dissociation constant of the base.



b

1.6.5 Self ionization of water;



$K_w = [\text{H}_3\text{O}^+][\text{OH}^-]$, where K_w is the ionic product of water.

$K_w = 10^{-14}$ at 298K when concentration is expressed in mol dm^{-3} . For a neutral solution,

$[\text{H}_3\text{O}^+] = [\text{OH}^-]$ and since $[\text{H}_3\text{O}^+][\text{OH}^-] = 10^{-14}$

$$[\text{H}_3\text{O}^+] = [\text{OH}^-] = 10^{-7} \text{ mol dm}^{-3}.$$

1.6.6 The pH scale

pH is the negative logarithm of the hydrogen (or hydroxonium ion) concentration.

Thus,

$$\text{pH} = -\log [\text{H}_3\text{O}^+]$$

In any solution, $[\text{H}_3\text{O}^+] [\text{OH}^-] = K_w$,

$$7$$

So that, $\log [\text{H}_3\text{O}^+] + \log [\text{OH}^-] = \log K_w$,

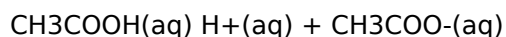
$$-\log [\text{H}_3\text{O}^+] - \log [\text{OH}^-] = -\log K_w$$

$$\text{pH} + \text{pOH} = \text{p}K_w = 14$$

In a neutral solution therefore, $\text{pH} = \text{pOH} = 7$.

1.6.7 Relationship between acids - bases ionization constants

Consider the following reaction;



$$K =$$

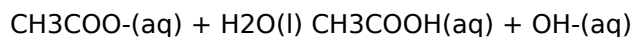
$$\frac{[\text{H}^+][\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]}$$

$$3$$

$$a$$

$$[\text{CH}_3\text{COOH}]$$

The conjugate base CH_3COO^- , reacts with water according to the equation



$$K_b =$$

$$\frac{[\text{CH}_3\text{COOH}][\text{OH}^-]}{[\text{CH}_3\text{COO}^-]}$$

$$3$$

$$3$$

Multiplying K_a by K_b gives,

$$K_a \cdot K_b =$$

$$\frac{[\text{H}^+][\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]}$$

$$\frac{[\text{CH}_3\text{COOH}][\text{OH}^-]}{[\text{CH}_3\text{COO}^-]}$$

$$[\text{H}^+][\text{OH}^-]$$

$$[\text{CH}_3\text{COOH}]$$

$$\times$$

$$\frac{[\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COO}^-]}$$

$$= [\text{H}^+][\text{OH}^-]$$

$$= K_w$$

Therefore,

$$K_a \cdot K_b = K_w$$

The above equation can be expressed as;

$$K_a = \frac{K_w}{K_b}$$

$$K_a = \frac{K_w}{K_b}, K_b = \frac{K_w}{K_a}$$

$K_b K_a$

This means that the stronger the acid (larger K_a), the weaker its conjugate base (smaller K_b) and vice versa.

The equation can also be expressed in terms of the logarithms.

$$\log K_a + \log K_b = \log K_w$$

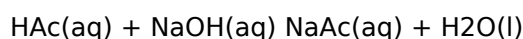
$$-\log K_a - \log K_b = -\log K_w$$

$$p K_a + p K_b = p K_w$$

A large value of $p K_a$ (small K_a) indicates a weak acid. The neutralization reaction between a strong acid and a strong base gives rise to a stable product. On the other hand, if the reaction is between a strong acid and a weak base, or a weak acid and a strong base, or a weak acid and a weak base, the product reacts with the solvent.

1.6.8 Salt of a weak acid and a strong base, e.g. NaAc.

In the reaction,



The product sodium ethanoate, NaAc reacts with water as follows;



$$K_h =$$

$\frac{[\text{HAc}][\text{OH}^-]}{[\text{Ac}^-]}$, where K is referred to as the hydrolysis constant.

$$[\text{Ac}^-]$$

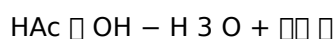
$$-h$$

8

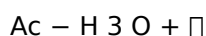
Multiplying both the numerator and denominator by $[\text{H}_3\text{O}^+]$, gives

$$K_h =$$

$$\frac{[\text{HAc}][\text{OH}^-]}{[\text{Ac}^-]}$$



$$\frac{[\text{HAc}][\text{OH}^-]}{[\text{Ac}^-]}$$



$$K$$

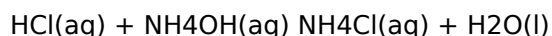
$$= \frac{[\text{HAc}][\text{OH}^-]}{[\text{Ac}^-]}$$

$$K_a$$

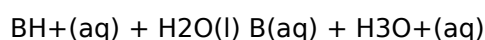
For a weak acid, K_a is small so that K_h will be large.

1.6.9 Salt of a strong acid and a weak base, e. g. NH_4Cl .

Consider the reaction;



The product NH_4Cl may be written as BH^+ and reacts with water as follows:



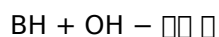
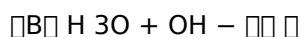
$$K_h =$$

$$\frac{[\text{B}][\text{H}_3\text{O}^+]}{[\text{BH}^+]}$$

$$[\text{BH}^+] +$$

Multiplying the numerator and denominator by $[\text{OH}^-]$, gives

$K_h =$

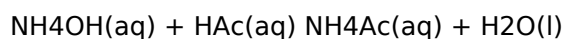


K

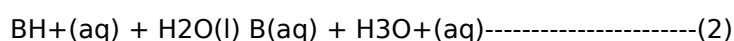
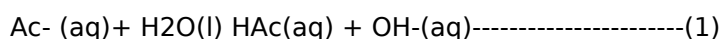
$= w$

K_b

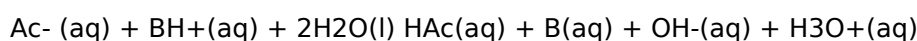
1.6.10 Salt of a weak acid and a weak base e. g NH_4Ac



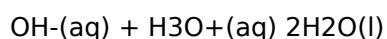
Writing NH_4Ac as BH^+Ac^- , we have the following reaction



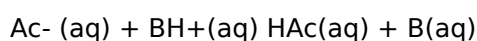
Combining (1) and (2) gives



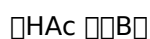
Now,



So that



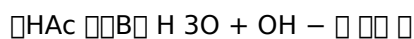
and $K_h =$



$= \frac{K_a}{K_b}$

Multiplying the numerator and denominator by $[H_3O^+][OH^-]$, gives

$K_h =$



K

$= w$

$K_a \cdot K_b$

9

1.7 Buffer solutions

A buffer solution is a solution of a weak acid or a weak base and its salt with the ability to resist changes in pH upon addition of small amounts of either acid or base.

A buffer solution must contain a relatively large concentration of acid to react with any OH^- ions that may be added to it and must contain similar concentration of base to react with any added H^+ ions. Furthermore, the acid and base components of buffer must not consume each other in a neutralization reaction. These requirements are satisfied by an acid-base conjugate pair (a weak acid and its conjugate base or a weak base and its conjugate acid)

A simple buffer solution can be prepared by adding comparable amounts of ethanoic acid (CH_3COOH) and sodium ethanoate (CH_3COONa) to water. The equilibrium concentrations of

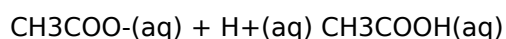
both the acid and the conjugate base (from CH_3COONa) are assumed to be the same as the starting concentrations. This is so because; (1) CH_3COOH is a weak acid and the extent of hydrolysis of CH_3COO^- ion is very small and (2) the presence of CH_3COO^- ions suppresses the ionization of CH_3COOH , and the presence of CH_3COOH suppresses the hydrolysis of CH_3COO^- ions.

A solution containing these two substances has the ability to neutralize either added acid or added base. Sodium ethanoate, a strong electrolyte, dissociates completely in water;

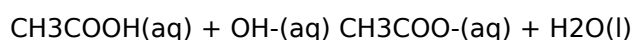
H_2O



If an acid is added, the H^+ ions will be consumed by the conjugate base in the buffer, CH_3COO^- , according to the equation;

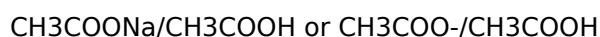


If a base is added to the buffer system, the OH^- ions will be neutralized by the acid in the buffer;



The buffering capacity, that is the effectiveness of the buffer solution, depends on the amounts of acid and conjugate base from which the buffer is made. The larger the amount, the greater the buffering capacity.

In general, a buffer system can be represented as salt/acid or conjugate base/acid. Thus the sodium ethanoate-ethanoic acid buffer system can be written as



1.7.1 Preparing a buffer solution with a specific PH. (The Henderson-Hasselbalch equation)

Referring to the ethanoic acid-sodium ethanoate buffer system, we can write the equilibrium constant as;

$$K_a =$$

$$\frac{[\text{CH}_3\text{COO}^-][\text{H}^+]}{[\text{CH}_3\text{COOH}]}$$



Rearranging the equation gives;

$$[\text{H}^+] = K_a$$



Taking the negative logarithm of both sides, we obtain

$$-\log [\text{H}^+] = -\log K_a - \log$$



$$\log$$

$$\frac{[\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]}$$



or

$$\log$$

$$-\log [\text{H}^+] = -\log K_a + \log$$



a

3

So that

$$\text{pH} = \text{pK}_a + \log$$



3



$$\text{pH} = \text{pK}_a + \log$$

a



This is called the Henderson-Hasselbalch equation.

Generally, it is written as

$$\text{pH} = \text{pK}_a + \log$$



Similarly for a mixture of a weak base and its salt with a strong acid

$$\text{pOH} = \text{pK}_b + \log$$



If the molar concentrations of the acid and its conjugate base are approximately equal, that is,



log



$$\text{Or } \text{pH} = \text{pK}_a$$

H

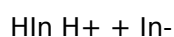
Thus, to prepare a buffer solution, we choose a weak acid whose pK_a is close to the desired pH .

This choice not only gives the correct pH value of the buffer system, but also ensures that we have comparable amounts of the acid and its conjugate base present; both are prerequisites for the buffer system to function effectively.

1.8 Acid-Base Indicators

These are weak organic acids or bases used for visual end point location in the course of a titration involving neutralization. Acid-base indicators change colour over a definite pH range and have different colours in acid and basic media. The difference in colours is as a result of differences in the colours of the conjugate acid-base forms.

Consider a weak acid indicator HIn . It dissociates as follows;



colour 1 colour 2

$K =$

$\frac{[H^+][In^-]}{[HIn]}$, where K_{In} is the dissociation constant of the indicator HIn .

$+ -$

In

$[HIn]$

Rearranging, we have;

$[H^+] = K \cdot \frac{[HIn]}{[In^-]}$

In^-

In

$pH = pK_{In} - \log$

$\frac{[Acid\ form]}{[basic\ form]}$

$\frac{[basic\ form]}{[acid\ form]}$

For a weak base indicator, $InOH$,

11

$InOH \rightleftharpoons In^- + OH^-$

$K =$

$\frac{[In^-][OH^-]}{[InOH]}$

$+ -$

In

$[InOH]$

Rearranging, we have;

$[OH^-]$

$[OH^-] = K \cdot \frac{[In]}{[InOH]}$

$[InOH]$

$[In] +$

$pOH = pK_{In} - \log$

$\frac{[Basic\ form]}{[Acid\ form]}$

$\frac{[Acid\ form]}{[Basic\ form]}$

Since indicators for acid-base titrations are weak organic acids and bases, and are added to the solution in very small amounts, the pH or pOH of the medium is essentially that of the solution. The colour changes depending on whether protons are removed or added to the solution.

1.9 Titration curves

A titration curve is a plot of pH versus the amount (usually volume) of acid or base added. It displays graphically the change in pH as acid or base is added to a solution and shows how pH changes near the equivalence point. The point at which the color of an indicator changes in a titration is known as the end point. It is determined by the K_{In} value for the indicator. Typically, color changes occur over a range of 1.5 to 2.0 pH units. The equivalence point is the point at which chemically equivalent amounts of acid and base have reacted. Ideally, the end point and the equivalence point in a titration should coincide. In practice, an indicator is selected such that

the range of color change includes the equivalence point.

1.9.1 Titration of a strong acid by a strong base (e.g HCl and NaOH)

1. Before any strong base is added, the pH depends on the strong acid alone.
2. After some strong base has been, but before the equivalence point, the remaining (excess) strong acid determines the equivalent pH.
3. At the equivalence point, the solution is neutral.
4. Beyond the equivalence point, excess strong base determines the pH.

12

1.9.2 Titration of a weak acid by a strong base (e.g CH₃COOH and NaOH)

1. Before any base is added, the pH depends on the weak acid alone.
2. After some base has been added, but before the equivalence point, a series of weak acid/salt (e. g. CH₃COOH/NaCH₃COO) buffer solutions determines the Ph.
3. At equivalence point, hydrolysis of the anions of the weak acid determines the pH.
4. Beyond the equivalence point, excess strong base determines the pH.

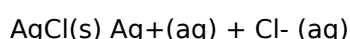
1.9.3 Titration of a weak acid by a weak base (e. g. CH₃COOH and NH₃)

In titration curves for weak acids and weak bases, pH changes near the equivalence point are too small for color indicators to be used. The solution is buffered both before and after the equivalence point.

13

1.10 Solubility product

Small amounts of sparingly soluble salts ionises in water. If a true equilibrium exists between the ions and the solid salt, an equilibrium constant expression can be written for the reaction. For example, silver chloride ionises as follows;



K=



+ -



At constant temperature, the concentration of AgCl is unchanged since it does not dissolve in the water. This constant can therefore be combined with the equilibrium constant, thus,



$K_{sp} = [\text{Ag}^+][\text{Cl}^-]$ or $K_{sp} = [\text{Ag}^+][\text{Cl}^-]$, where K_{sp} is known as the solubility product. It is the ionic product in a saturated aqueous solution at a given temperature. In an unsaturated solution, the ionic product is less than K_{sp} , therefore, any precipitate will dissolve until ionic product equals the value of K_{sp} . A super saturated solution contains more solute that can dissolve at a given temperature; the ionic product therefore, is greater than K_{sp} . Precipitation must take place in such a solution to make K_{sp} equal to the ionic product.

1.10.1 Application of solubility product

In both qualitative and quantitative chemical analysis, it may be necessary to remove one salt

from a solution by using differences in solubilities of the salts. Consider the separation of Zn^{2+} and Fe^{2+} as their sulphides. Their solubility products are;

$[\text{Zn}^{2+}][\text{S}^{2-}] = 4.5 \times 10^{-24}$ and $[\text{Fe}^{2+}][\text{S}^{2-}] = 1 \times 10^{-19}$, the ZnS being less soluble.

Suppose the Zn^{2+} and Fe^{2+} concentration is 0.1 mol dm^{-3} . On adding S^{2-} ions to the solution, the precipitation of FeS occur only when $[\text{Fe}^{2+}][\text{S}^{2-}] = K_{\text{sp}}$. That is, when

$$[\text{S}^{2-}] = \frac{K_{\text{sp}}}{[\text{Fe}^{2+}]}$$

$$= \frac{1 \times 10^{-19}}{0.1}$$

$$= 1.0 \times 10^{-18} \text{ mol dm}^{-3}$$

$$K_{\text{sp}} = 4.5 \times 10^{-24}$$

$$= \frac{4.5 \times 10^{-24}}{0.1}$$

$$= 4.5 \times 10^{-23}$$

$$[\text{S}^{2-}] = 4.5 \times 10^{-23}$$

$$[\text{S}^{2-}] = 4.5 \times 10^{-23}$$

While the precipitation of ZnS will occur when $[\text{S}^{2-}] = \frac{K_{\text{sp}}}{[\text{Zn}^{2+}]}$

$$= \frac{4.5 \times 10^{-24}}{0.1}$$

$$= 4.5 \times 10^{-23}$$

$$= 4.5 \times 10^{-23}$$

The $[\text{Zn}^{2+}]$ ion remaining in solution when FeS precipitates is

$$[\text{Zn}^{2+}] = \frac{K_{\text{sp}}}{[\text{S}^{2-}]}$$

$$= \frac{4.5 \times 10^{-24}}{4.5 \times 10^{-23}}$$

$$K_{\text{sp}} = 4.5 \times 10^{-24}$$

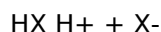
$$[\text{Zn}^{2+}] = \frac{4.5 \times 10^{-24}}{4.5 \times 10^{-23}} = 0.1 \text{ mol dm}^{-3}$$

$$[\text{S}^{2-}] = 4.5 \times 10^{-23}$$

This indicates that the precipitation of ZnS is essentially complete by the time FeS begins to precipitate and the two ions can be easily separated by precipitation method.

1.10.2 Common-Ion effect

Consider the equilibrium,



in a solution of HX. This equilibrium is affected when more H^+ or X^- are

added to the system from an external source. The system will not readily accept the added ions as it would if the same ions were not already present in the solution. This phenomenon is known as common ion effect. It affects the solubility of an electrolyte.

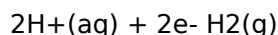
14

2.0 ELECTROCHEMISTRY

This branch of chemistry deals with the use of chemical reactions to produce electricity and the use of electricity to produce a chemical change. The process relies on redox reactions at the electrodes to produce ions in solution. These ions are responsible for maintaining electrolytic conduction; that is, the conduction of electric current by means of ions of electrolyte in solution or in molten state.

2.1 Electrode reactions

In an electrochemical cell, reduction occurs at the cathode and oxidation at the anode. Consider the electrolytic cell below; electrons enter the solution at the cathode, where H^+ or Na^+ ions react with them, leading to reduction of the cation



Electrons leave the solution at the anode by discharge of Cl^- or OH^- ions. The anions thus becomes oxidised.

Figure 1. An electrolytic cell.

The electrode reactions indicate that oxidation is a process of electron loss and reduction is a process of electron gain. Both processes occur simultaneously in the cell, since every electron lost at the anode must be gained at the cathode to maintain electrical neutrality of the electrolyte solution.

15

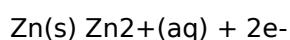
2.2 Electrochemical cells

The experimental apparatus for generating electricity through the use of a redox reaction is called an electrochemical cell.

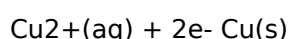
Fig 2. Daniel cell

A zinc bar is dipped into a ZnSO_4 solution and a bar of Cu is dipped into CuSO_4 solution. The galvanic cell operates on the principle that oxidation of Zn to Zn^{2+} and a reduction of Cu^{2+} to Cu can be made to take place simultaneously in separate locations with the transfer of electrons through the external wire. As the reaction progresses, it sets up a constant flow of electrons and hence, generates electricity. A salt bridge which is a glass tube connecting concentrated KCl or KNO_3 solution joins the two compartments together and serves to keep the two solutions electrically neutral. The origin of the current is through the half cell reactions at the electrodes.

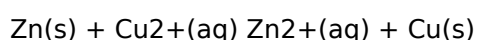
Anode;



Cathode;



Overall reaction;



2.3 Electrode potential (E)

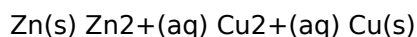
This is the tendency of an electrode to release to or attract electron from the surrounding electrolyte solution. The zinc plate in the example above has a negative electrode potential and thus, a great tendency to lose electrons to the surrounding solution. Another way of describing the phenomenon is to say that zinc in contact with a molar solution of its own ions has a greater electrolytic solution pressure than deposition pressure. Copper on the other hand has a positive electrode potential or greater deposition pressure.

16

2.4 Cell representation

The arrangement of electrodes in a cell can be summarized by the notation such as that shown

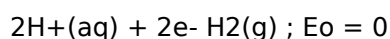
below for the Daniel cell.



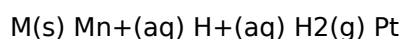
The cathode is shown on the right hand and the anode on the left. The double vertical bar separating the two compartments represents the salt bridge.

2.5 Standard electrode potential (E°)

This is the potential measured under standard conditions of temperature (298K), pressure (1 atmosphere) and concentration (1 mol dm⁻³) of metal ion solution. It is measured against a reference standard. The hydrogen electrode has been adopted as the standard and it consists of H₂ gas at 1 atmosphere pressure, in contact with a molar solution of its ions at 298K. A platinum electrode coated with platinum black is incorporated to catalyse the attainment of equilibrium between H₂ gas and H⁺ ions in the solution.



The hydrogen electrode under these conditions is assigned a value of zero by convention. The standard electrode potential of a given system $\text{Mn}^+(\text{aq}) \mid \text{M(s)}$ is found by connecting it to a H₂ electrode via a salt bridge. By convention, the negative pole is assigned a negative potential and the positive pole, a positive potential.



$E^\circ(\text{M})$ $E^\circ(\text{SHE})$

17

Figure 3. Set up for the standard hydrogen electrode

A table of standard hydrogen electrode potential is similar to the electrochemical series. Metals higher than hydrogen in the series (with alkaline metals lower down have a positive electrode potential while those below has a negative electrode potential (also called redox potential). In other words, the lower the position of a couple in the table, the greater the reducing power of the reduced substance. It follows that when two couples form a cell, the couple lower down in the table form the anode of the cell and the couple higher up forms the cathode. Thus, in the half cell reaction, the lower couple can be written as the oxidation and the upper couple as a reduction.

Lower couple upper couple

Anode compartment Cathode compartment



2.6 Dependence of redox potentials on ionic concentration and temperature.

The value of the potential difference between a metal and a solution of its own ion is dependent upon the ionic concentration and the temperature. If the ionic concentration is decreased, the tendency for the metal ions to deposit on the metal will be reduced, that is, the electrode potential of the metal will become less positive. Similarly, if the metal ion concentration is increased, the electrode potential will be more positive. The dependence of electrode potential on metal ion concentration and temperature is expressed by the Nernst equation,

18

$$E = E^{\circ} + \frac{RT}{nF} \ln M_n +$$

Where E is the e.m.f of the cell and E° is the standard electrode potential, s.e.p, (for a molar solution of metal ions at 298K. R is the gas constant (8.31 Jmol⁻¹ K⁻¹), T is the temperature in Kelvin, n is the number of electrons transferred and F is the Faraday.

Verification of the equation involves the measurement of electrode potential of a metal for different metal ion concentrations at a given temperature. A metal (metal ion half-cell) is connected via a salt bridge to a standard hydrogen electrode. The potential of various cells are recorded and a plot of E values against $\ln[M_n^{+}]$ gives a straight line. A similar graph is obtained at various temperatures for fixed concentrations of M_n^{+} .

2.7 Cell potential and reaction free energy

Cell potential, E, is proportional to the free energy of reaction.

$$\Delta G^{\circ} = -nFE^{\circ}$$

This is the case since a large negative value of ΔG indicates that the reaction has a tendency to occur spontaneously and drive electrons through the circuit. If ΔG is negative but small, we expect a small cell potential, while the value of $\Delta G = 0$ indicates that the cell reaction is at equilibrium and has no tendency to drive electrons through the circuit. Under this condition $E=0$ and no current flows in the circuit. The chemical energy released is equal to the decrease in free energy. If n moles of electrons are transferred per mole of chemical energy, the decrease in free energy is

$$-\Delta G = nFE \text{ or } \Delta G = -nFE$$

where F is the faraday (96500 Coulombs).

If the constituents of the cell are in their standard states, then we define the standard free energy of the half-cell reaction as

$$\Delta G^{\circ} = -nFE^{\circ}$$

where E° is the standard electrode potential and ΔG° is the standard free energy of the reaction. This is a measure of the tendency of reactants in their standard states to yield products in their standard states.

2.8 Equilibrium constants of cell reactions

Since

$$\Delta G^{\circ} = -nFE^{\circ}$$

$$\Delta G^{\circ} = -RT \ln K \text{ and } \Delta G^{\circ} = -nFE^{\circ}, \text{ then } \ln K =$$

$$\frac{nFE^{\circ}}{RT}$$

The equilibrium constant for a cell reaction can thus be calculated from the above relationship.

19

RATES OF CHEMICAL REACTION

1.1 CHEMICAL KINETICS

- Chemical kinetics:

– Study of rate of change of concentrations of substances involved in chemical reactions

- Reaction rate:

– How rapidly a reaction occurs

– Related to rates of change in concentration of reactants and products over time

- Factors affecting reaction rates:

– Physical state of reactants

– Concentration of reactants

– Temperature

– Presence of a catalyst

– Physical state of the reagent

– Molecular orientation

The amount of NO formed in a given period of time depends on how fast the reaction proceeds.

– We can define the rate of the reaction in terms of the change in NO over time. The change in concentration can be

calculated as the final concentration (at some time) minus the initial concentration. Since NO is a product, its

concentration is increasing over time and so the rate is positive.

– Alternatively, we can express the reaction rate in terms of how fast a given reactant is consumed, such as the

consumption of N₂ and O₂ are shown on the slide. In all cases the rate is expressed in terms of a change in

concentration of a reactant or product (in units of moles/L) as a function of time as in the equation

Suppose we ran a reaction in the laboratory to determine the rate of the reaction between N₂ and O₂. We can express

the rate either in terms of the rate of consumption of N₂ or O₂, or in terms of the rate of formation of NO.

– A typical set of concentration data for these species are presented in Table 13.1 and are graphed in Figure 13.6. As

expected, the concentration of reactants decreases and the concentration of product increases as the reaction

proceeds.

– If we look at the change in concentration of NO between 5 and 10 μ sec, we can calculate a rate by noting the change

in concentration over this time interval, and obtain a rate of 1.4 M/s.

$\Delta[\text{NO}]$

Average rate =

Δt

$14.8 \mu\text{M} - 7.8 \mu\text{M}$

Average rate = $10 \mu\text{s} - 5 \mu\text{s} = 1.4 \text{ M/s}$ ($\text{M} = \mu\text{M/L}$)

CHM111_2020_Dr salihu 1

Similarly, if we define the rate in terms of the change in [N₂] over the same

time interval, we calculate a rate of 0.70 M/s. Note that this is half the rate of formation of product NO, as expected based on the reaction stoichiometry.

The reaction rates calculated in this manner represent average reaction rates (Figure 1), and are only valid for the time interval in question. Any other time interval (e.g., between 25 Figure 1 and 30 s) will yield a different value.

Average rate of a reaction is the change in concentration of reactant or product over a specific time interval

$$\Delta[\text{O}_2]$$

$$\text{Average rate} = -$$

$$\Delta t$$

$$(9.6 \mu\text{M} - 13.1 \mu\text{M})$$

$$\text{Average rate} = - \frac{10 \mu\text{M} - 5 \mu\text{M}}{5 \text{ s}} = -(-0.7) = 0.7 \text{ M/s} \quad (\text{M} = \mu\text{M/L})$$

The instantaneous rate of a reaction can also determine as the tangential slope of the concentration plotted versus time. This is illustrated on the slide for the reaction between NO and O₂ to form NO₂.

The change in concentration of O₂ over a specific time interval can be used to calculate the average rate of consumption of O₂ during that time interval, but can also be used to determine the instantaneous rate at the midpoint of that time interval.

Figure 2

CHM111_2020_Dr salihu 2

Instantaneous rate: Reaction rate at a particular instant, it is determined graphically as tangential slope of

concentration versus time plot. The Figure 2 is to determine the rate of the reaction at 2000.0 s. This instantaneous rate can be seen more clearly as the tangential slope in this expanded section of the [O₂] curve (Figure 3). Using the change in O₂ concentration between 1000 and 3000 sec, we can calculate the average rate of consumption of O₂ as $6.0 \times 10^{-7} \text{ M/s}$. This calculated rate represents the average rate over the time interval from 1000 to 3000 sec, but also represents the instantaneous rate at the midpoint of this time

Figure 3

interval (at 2000 sec).

Initial rate: Consider the concentration data plotted in Figure 4. The tangential slope of the curve at points b and c

represent instantaneous rates (tangential slopes) at specific times in the reaction. The slope at point a when $t = 0$

represents what is known as the initial rate.

- Initial rate: the rate of a reaction at $t = 0$, immediately after the reactants are mixed

Clearly the reaction rate (i.e., tangential slope of the concentration curve) decreases as the reaction proceeds.

Eventually, we can imagine that the rate of change would approach zero, at which point concentrations of reactants and products no longer change.

This behavior is typical for most reactions. The most rapid reaction rates occur early in the reaction and reaction rates decrease over time. We can explain this in terms of molecular collisions. As concentrations decrease over time, the number of collisions between reacting species decreases and the reaction rate slows down accordingly.

Figure 4

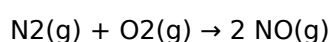
Exactly how reaction rates depend on concentrations of reactants, however, is not always obvious. If we double the concentration of one reactant, the reaction rate may also

double. If we double the concentration of another reactant, however, the reaction rate might quadruple. We need a

way to determine the nature of the dependency of rates on concentrations of reactants.

1.1.1 PRACTICE

(i) Calculate the average reaction rate for the time period between 15 μ s and 30 μ s using the data below:



Time [N₂], [O₂] [NO] (μ M)

(μ s) (μ M)

0 17.0 0.0

15 7.6 18.6

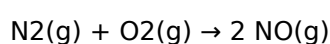
30 3.6 26.7

CHM111_2020_Dr salihu 3

Collect and Organize: We are given the balanced chemical equation, as well as concentrations of reactants and

products over time. We are asked to calculate the average rate of reaction between 15 and 30 μ s.

(ii) Calculate the average reaction rate for the time period between 15 s and 30 s using the data below:



Time [N₂], [O₂] [NO]

(s) (M) (M)

0 17.0 0.0

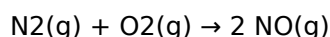
15 7.6 18.6

30 3.6 26.7

Analyze: Using the data from the table, we can calculate the rate either as the rate of formation of NO over the stated

time interval or as the rate of consumption of N₂ or O₂. Reaction rate is defined as change in concentration over time.

(iii) Calculate the average reaction rate for the time period between 15 s and 30 s using the data below:



Time (s) [N₂], [O₂] (M) [NO] (M)

0 17.0 0.0

15 7.6 18.6

30 3.6 26.7

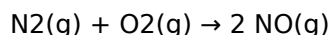
Solve: If we define the rate in terms of formation of NO we obtain

$$\text{Rate} = - ([26.7 - 18.6\text{M}]/(30 - 15\text{s})) = 5.4 \times 10^5 \text{ M/s}$$

If we define the rate in terms of consumption of N₂ or O₂ over the same interval we obtain

$$\text{Rate} = -([3.6 - 7.6\text{M}]/(15\text{s})) = 2.7 \times 10^5 \text{ M/s}$$

(iv) Calculate the average reaction rate for the time period between 15 s and 30 s using the data below:



Time (s) [N₂], [O₂] (M) [NO] (M)

0 17.0 0.0

15 7.6 18.6

30 3.6 26.7

Note: The rate of formation of product NO is twice the rate of consumption of reactants (N₂, O₂) over the same interval,

consistent with the stoichiometry of the balanced chemical equation.

CHM111_2020_Dr salihu 4

1.2 RATE LAWS AND REACTION ORDER

Experimental observations and theoretical considerations tell us that the reaction rate depends on reactant

concentrations, but they do not tell us the exact dependency. Based on experimental data we can determine the nature

of this dependency and express it as reaction order, a parameter that tells us how a reaction rate depends on reactant

concentration.

Knowing the reaction order also provides insight into how a reaction might proceed on the molecular level. For

example, which molecules must collide and which bonds must break in order to form the observed products.

We can summarize the relationship between reaction rates and the concentrations of reactants using a rate law, which

is a mathematical relationship that defines the experimentally observed relationship between rates and concentrations

(i.e., reaction orders).

We can also determine the overall order of the reaction as the sum of the reaction orders of the individual reactants.

- Rate law:

- Equation that defines the experimentally determined relationship between reactant concentrations and

the rate of the reaction

- Reaction order:

- Experimentally determined number defining the dependence of the reaction rate on the concentration

of a reactant

- Overall order of reaction: sum of exponents of concentration terms in the rate law

Included in the rate law is a proportionality constant, k , called the rate constant that relates the concentration of

reactants to the observed reaction rate.

- Consider a generic reaction: $A + B \rightarrow C$

- We can write a generic rate law for this reaction as:

$$\text{Rate} = k[A]^m[B]^n$$

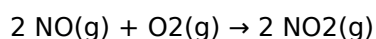
Where the rate is defined in terms of the change in concentration of a given species over time (can be initial rate or

instantaneous rate), k is the rate constant, $[A]$ and $[B]$ are the concentrations of reactants at a given time, and m and

n are the reaction orders with respect to the reactants A and B . The values of m , n , and k must be determined

experimentally.

1.2.1 DETERMINING RATE LAWS



- Rate = $k[\text{O}_2]^m[\text{NO}]^n$

- Determine values of n and m by measuring initial rate under different conditions.

CHM111_2020_Dr salihu 5

Figure 5

Consider the reaction data in Table of Figure 5. Note that the initial concentration of NO is the same in trials 1 and 2,

but the concentration of O₂ changes by a factor of two. Any change in the rate of the reaction in this case must be due

to the change in [O₂].

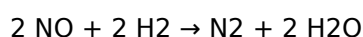
Similarly, the initial [O₂] remains the same between trials 1 and 3, but the initial [NO] changes by a factor of 2. Any

change in the reaction rate in this case must be due to the change in [NO]. We can use these data to mathematically

determine the reaction order with respect to [NO] and [O₂].

- Rate = $k[\text{O}_2]^m[\text{NO}]^n$
- To find m, Compare experiments #1 and #2.
- [NO] is constant.
- [O₂] is halved.
- Rate is halved.
- Rate $\propto [\text{O}_2]^m$
- $m=1$
- To find n, Compare experiments #1 and #3.
- [O₂] is constant.
- [NO] decreases by x2.
- rate decreases by x4.
- Rate $\propto [\text{NO}]^2$
- $n=2$
- Rate = $k[\text{O}_2][\text{NO}]^2$
- k can be calculated from the rate law using data for any experiment.

□ Write the rate law and calculate the rate constant k for the reaction given, using the following data:



Trial [NO], M [H₂], M Init. Rate, M/s

-5

1 0.00570 0.140 7.01×10

-4

2 0.00570 0.280 1.40×10

-4

3 0.0114 0.140 2.81×10

CHM111_2020_Dr salihu 6

Solution:

- Write the generic rate law is: Rate = $k[\text{NO}]^m[\text{H}_2]^n$.
- The actual values for m and n must be determined mathematically from the experimentally observed reaction rate data. Once the reaction orders are determined, data for any trial can be inserted into the rate law to calculate k.

- Compare trials 1 and 2: the $[H_2]$ doubles while $[NO]$ is held constant, and the rate also doubles. The reaction

order with respect to H_2 is $n = 1$.

- Compare trials 1 and 3: the $[H_2]$ is constant while the $[NO]$ is doubled. The rate quadruples, so the reaction order

with respect to $[NO]$ is $m = 2$.

- The rate law is now written as: $\text{Rate} = k[NO]^2[H_2]$

- Overall reaction order = 3.

- Substituting data for trial #1: $7.01 \times 10^{-5} \text{ M/s} = k(0.00570)^2(0.140)$

- Rearranging and solving for k :

- $k = (7.01 \times 10^{-5} \text{ M/s}) / [(0.00570 \text{ M})^2(0.140 \text{ M})] = 15.4 \text{ M}^{-2}\text{s}^{-1}$

1.2.2 INTEGRATED RATE LAWS

1.2.2.1 FIRST ORDER

Determining rate laws using initial rates has two disadvantages:

1. multiple experiments must perform with varying initial concentrations.
2. the initial rate for each experiment must be determined accurately.

Therefore, it would be faster and easier if we could determine the rate law and rate constant by monitoring

concentrations for one experiment. This is possible using the integrated rate law approach.

- Consider the decomposition of ozone $O_3 \rightarrow O_2 + O$ catalysed by sunlight

Since there is only one reactant, we can write a generic rate law as $\text{rate} = -\Delta[O_3]/\Delta t = k[O_3]^n$. If we assume that the

reaction is first order, then $n = 1$.

- If we rearrange the rate law, isolating the concentration terms on one side and the $k\Delta t$ term on the other side, we can

apply integral calculus to obtain: $\ln[O_3]_t = -kt + \ln[O_3]_0$.

- This integrated rate law takes the form of an equation for a straight line: $y = mx + b$,

where $y = \ln[O_3]$, $m = \text{slope } (-k)$, $x = \text{time}$,

and b (intercept) = the initial concentration of ozone, $[O_3]_0$. If the reaction is first order,

then plotting $[O_3]$ versus time will yield a

straight line with slope = $-k$.

In a simplified form:

1.2.2.2 GRAPHICAL DETERMINATION OF REACTION ORDER

Typical data for the photodecomposition of ozone over time are presented in the following Table and Figure, along with

a plot of concentration versus time. This plot shows a curved line with changing rates over time. It is difficult to

determine the rate constant for the reaction from this plot.

CHM111_2020_Dr salihu 7

Figure 6

Typical data for the photodecomposition of ozone over time are presented in the table of Figure 6, along with a plot of the log of concentration versus time. Consistent with the expectations for a first order reaction, this plot produces a straight line.

- Using the data, we can calculate the slope as $\Delta \ln[\text{O}_3]/\Delta t = -k = -1.1 \times 10^{-3} \text{ s}^{-1}$

$\Delta \ln[\text{O}_3]$

Slope = $-k$

Δt

$= -1.1 \times 10^{-3}$

$-3 -1$

$= -k$

1.2.3 SECOND ORDER

The decomposition of nitrogen dioxide has been studied as found to be 2nd order with respect to the reactant: NO₂

$\text{NO} + \text{O}_2$ and the rate law of rate is given by

$\Delta[\text{NO}_2]$

rate law of the rate = $k[\text{NO}_2]^2$

Δt

- Following the same logic as for the first-order rate law, we rearrange the rate law equation and integrate with respect

to time (Δt), which produces an integrated rate law of

$\frac{1}{[\text{NO}_2]_t}$

$= kt +$

$\frac{1}{[\text{NO}_2]_0}$

Again, this equation takes the form of a straight line. Plotting $1/[\text{NO}_2]_t$ vs time produces a linear plot Figure 7 (b) with

a slope = k and an intercept = $1/[\text{NO}_2]_0$.

$2\text{NO}_2(\text{g}) \rightarrow 2\text{NO}(\text{g}) + \text{O}_2(\text{g})$

Simplified form

CHM111_2020_Dr salihu 8

Figure 7

1.2.4 ZERO ORDER REACTION

It is also possible that the rate of a reaction may be independent of the concentration of a given reactant. Such reactions

are called zero-order reaction with regard to that reactant

1.3 FACTORS AFFECTING RATES OF CHEMICAL REACTION

(i) Temperature (T)

As molecules collide, bonds in reactants are broken which allows new bonds to form to yield the products. The collisions

between molecules must be of sufficient energy to break the bonds in reactants or no reaction occurs. The temperature

of a reaction will affect the energy of the collisions because the kinetic energy of molecules is directly related to

temperature. As T increases we would expect reaction rate to increase since kinetic energy of the molecules increases.

Note: Increased temperature increases kinetic energy of molecules and molecular collisions

- The minimum amount of collision energy required for reaction is called the activation energy (E_a).

Every reaction

has a characteristic activation energy, expressed in kJ/mol.

(ii) Activation energy (E_a):

The minimum energy of molecular collisions required to break bonds in reactants, leading to formation of products

These factors (T and E_a) are reflected in the rate law by incorporation into the rate constant.

$$k = A e^{-E_a/RT}$$

R = Universal gas constant in J/mol./K; T = Temperature (kelvin); A is a collisional frequency factor

CHM111_2020_Dr salihu 9

$$-E_a/RT$$

$$\ln k = \ln A - E_a/RT$$

$$1/T$$

Reaction profile :

Figure 8

From the

Figure 8, note the relative internal energy of the reactants on the left and the internal energy of the products on the right.

As the reaction proceeds, collisions occur between reactant molecules. If a collision has sufficient energy, it produces

an intermediate state between reactants and products, called an activated complex. The energy difference between

the original reactants and the activated complex is the activation energy. Collisions having sufficient energy to

overcome this energy barrier and form the activated complex will yield products.

Note that the bigger the E_a , the fewer collisions you would expect to have sufficient energy and the slower the reaction

rate (i.e., the smaller the value of k); hence, the negative exponential dependence of k on E_a in the Arrhenius equation.

(iii) Effect of concentration

We can rationalize the dependence of reaction rates on concentrations by considering the illustration in Figure 9 for

the reaction between NO and O₃. The rate law indicates that the rate is first order with respect to NO and O₃.

- For a reaction to occur there must be a collision between reacting species (NO and O₃). In the first flask, the rate depends on the collisions as indicated

CHM111_2020_Dr salihu 10

Figure 9

. - Moving to the second flask, the concentration of NO has doubled, so the probability of a collision between an NO

and an O₃ molecule also doubles.

- Moving to the third flask, we now double the concentration of O₃. The probability of collisions increases by another

factor of two, or is now four times greater than in the first flask.

- Similarly, increasing the concentration of NO and O₃ by a factor of three each, compared to the first flask, increases

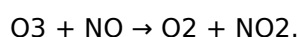
the probability of a collision by 32 or by a factor of nine.

(iv) Effect of molecular orientation

To understand the role of collision orientation

and its contribution to the frequency factor,

consider the reaction:



In the reaction sequence on the left Figure

10(A), the O₃ and NO molecules collide in an

orientation that allows one of the O atoms to

form a bond with the N atom in NO, forming

Figure 10 the activated complex and leading to

products.

In the reaction sequence on the right, Figure 10(B) the reactants are oriented so that the O atom in NO is in line with

the O₃. Even if the collision energy is sufficient to break the O₃ bonds, the free O atom is not in position to form a bond

with the N atom in NO, and so the activated complex is not formed and no reaction occurs.

- This dependence on collision orientation is reflected in the A constant and depends on the probability of a collision

occurring in an appropriate orientation to yield products.

(v) Effect of pressure

The effect of pressure on the reaction rate is pronounced in reaction involving gas phase. The molecules in bottle (B)

will have little space and may collide more frequently with each other, Figure 11. Therefore, the reaction in B will be

faster.

CHM111_2020_Dr salihu 11

Figure 11

(vi) Effect of physical state of the reactants

Physical state of reactants—in general, reaction rates are faster for substances in the liquid or gas phase than in the solid state. Even in the solid state, reaction will be faster in the powdered forms than the lump forms because the surface area in the powdered state is larger and hence contact between molecules or atoms of the reactants increases.

(vii) Effect of catalyst

As stated earlier, for a reaction to occur, the reactants collided with each other. The effective collision yield an energy equal or greater than the activation energy. The catalyst creates a reaction route with smaller E_a . Thus the reaction proceed faster than usual as shown as follow (Figure 12):

Figure 12

Catalysts can be classified as homogeneous and heterogeneous. Homogeneous catalysts are the ones in the same phase as the reactants, while the catalysts in a different phase with the reactants are called heterogeneous catalyst.

Catalysts can also be biological (e.g enzymes) or non-biological or chemical, eg the surface of a metal.

CHM111_2020_Dr salihu 12

COLLISION THEORY

The kinetic molecular theory of gases postulates that gas molecules frequently collide with one another.

Therefore, it seems logical generally true that chemical reactions occur as a result of collisions between reacting molecules.

In terms of the collision theory of chemical kinetics, then, we expect the rate of a reaction to be directly

proportional to the number of molecular collisions per second, or to the frequency of molecular collisions:

Rate of reaction $\propto$ Number of Collisions per second

Generally, any condition or factor that increase the number of collision will definitely increase the rate of reaction.

The implication of the collision theory is

- i. that a reaction always occurs when an A and a B molecule collide.
 - ii. The collision must be effective
 - iii. The energy of the collision must be equal or greater than the activated energy.
 - iv. The molecules must collide in particular orientation that ensure maximum contact and impact
- When molecules collide, part of their kinetic energy is converted to vibrational energy. If the initial kinetic

energies are large, then the colliding molecules will vibrate so strongly as to break some of the chemical

bonds. This bond fracture is the first step toward product formation. If the initial kinetic energies are small,

the molecules will merely bounce off each other intact.

Potential energy profiles for (a) exothermic and (b) endothermic reactions. These plots show the change in potential energy as reactants A and B are converted to products C and D. The activated complex ($AB^\ddagger$) is a highly unstable species with a high potential energy. The activation energy is defined for the forward reaction in both (a) and (b). Note that the products C and D are more stable than the reactants in (a) and less stable than those in (b)

We postulate that in order to react, the colliding molecules must have a total kinetic energy equal to or

greater than the activation energy (E_a), which is the minimum amount of energy required to initiate a

chemical reaction. When molecules collide they form an activated complex (also called the transition state),

CHM111_2020_Dr salihu 13

a temporary species formed by the reactant molecules as a result of the collision before they form the

product

The activation energy as a barrier that prevents less energetic molecules from reacting.. Because the

number of reactant molecules in an ordinary reaction is very large, the speeds, and hence also the kinetic

energies of the molecules, vary greatly. Normally, only a small fraction of the colliding molecules—the

fastest-moving ones—have enough kinetic energy to exceed the activation energy. These molecules can

therefore take part in the reaction

SUMMARY

1. The rate of a chemical reaction is the change in the concentration of reactants or products over time. The

rate is not constant, but varies continuously as concentrations change.

2. The rate law expresses the relationship of the rate of a reaction to the rate constant and the concentrations

of the reactants raised to appropriate powers. The rate constant k for a given reaction changes only with

temperature.

3. Reaction order is the power to which the concentration of a given reactant is raised in the rate law. The

order gives how large (magnitude) the concentration of the given reactant affects/depends on the rate of the

reaction. Overall reaction order is the sum of the powers to which reactant concentrations are raised in the

rate law. The rate law and the reaction order cannot be determined from the stoichiometry of the overall

equation for a reaction; they must be determined by experiment. For a zero-order reaction, the reaction rate

is equal to the rate constant.

4. The half-life of a reaction (the time it takes for the concentration of a reactant to decrease by one-half)

can be used to determine the rate constant of a first-order reaction.

5. In terms of collision theory, a reaction occurs when molecules collide with sufficient energy, called the

activation energy, to break the bonds and initiate the reaction. The rate constant and the activation energy

are related by the Arrhenius equation.

6. The overall balanced equation for a reaction may be the sum of a series of simple reactions, called

elementary steps. The complete series of elementary steps for a reaction is the reaction mechanism.

7. If one step in a reaction mechanism is much slower than all other steps, it is the rate-determining step.

8. A catalyst speeds up a reaction usually by lowering the value of E_a . A catalyst can be recovered unchanged at the end of a reaction.

9. In heterogeneous catalysis, which is of great industrial importance, the catalyst is a solid and the reactants

are gases or liquids. In homogeneous catalysis, the catalyst and the reactants are in the same phase. Enzymes

are catalysts in living systems.

References

1. Raymond Chang. Chemistry. 10th edition. McGraw-Hill, NY, 2010

2. Gilbert, Kirss, Foster, Bretz and Davies, Chemistry. 5th edition, W.W. Norton & Co 2017

CHM111_2020_Dr salihu 14

CHM111_2021Energy

Types of Energy

The various types of energy include kinetic, potential, and chemical energy.

Learning Objectives

Differentiate between types of energy

Key Takeaways

Key Points

□ All organisms use different forms of energy to power the biological processes that allow them to grow and survive.

□ Kinetic energy is the energy associated with objects in motion.

□ Potential energy is the type of energy associated with an object's potential to do work.

□ Chemical energy is the type of energy released from the breakdown of chemical bonds and can be harnessed for metabolic processes.

Key Terms

□ chemical energy: The net potential energy liberated or absorbed during the course of a chemical reaction.

□ potential energy: Energy possessed by an object because of its position (in a gravitational or electric field), or its condition (as a stretched or compressed spring, as a chemical reactant, or by having rest mass).

□ kinetic energy: The energy possessed by an object because of its motion, equal to one half the mass of the body times the square of its velocity.

Energy is a property of objects which can be transferred to other objects or converted into different forms, but cannot be created or destroyed. Organisms use energy to survive, grow, respond to stimuli, reproduce, and for every type of biological process. The potential energy stored in molecules can be converted to chemical energy, which can ultimately be converted to kinetic energy, enabling an organism to move. Eventually, most of energy used by organisms is transformed into heat and dissipated.

Kinetic Energy

Energy associated with objects in motion is called kinetic energy. For example, when an airplane is in flight, the airplane is moving through air very quickly—doing work to enact change on its surroundings. The jet engines are converting potential energy in fuel to the kinetic energy of movement. A wrecking ball can perform a large amount of damage, even when moving slowly. However, a still wrecking ball cannot perform any work and therefore has no kinetic energy. A speeding bullet, a walking person, the rapid movement of molecules in the air that produces heat, and electromagnetic radiation, such as sunlight, all have kinetic energy.

Potential Energy

What if that same motionless wrecking ball is lifted two stories above a car with a crane? If the suspended wrecking ball is not moving, is there energy associated with it? Yes, the wrecking ball has energy because the wrecking ball has the potential to do work. This form of energy is called potential energy because it is possible for that object to do work in a given state.

Objects transfer their energy between potential and kinetic states. As the wrecking ball hangs motionlessly, it has 0%0% kinetic and 100%100% potential energy. Once the ball is released, its kinetic energy increases as the ball picks up speed. At the same time, the ball loses potential energy

as it nears the ground. Other examples of potential energy include the energy of water held behind a dam or a person about to skydive out of an airplane.

Potential energy vs. kinetic energy: Water behind a dam has potential energy. Moving water, such as in a waterfall or a rapidly flowing river, has kinetic energy.

Chemical Energy

Potential energy is not only associated with the location of matter, but also with the structure of

matter. A spring on the ground has potential energy if it is compressed, as does a rubber band that is pulled taut. The same principle applies to molecules. On a chemical level, the bonds that hold the atoms of molecules together have potential energy. This type of potential energy is called chemical energy, and like all potential energy, it can be used to do work.

For example, chemical energy is contained in the gasoline molecules that are used to power cars. When gas ignites in the engine, the bonds within its molecules are broken, and the energy released is used to drive the pistons. The potential energy stored within chemical bonds can be harnessed to perform work for biological processes. Different metabolic processes break down organic molecules to release the energy for an organism to grow and survive.

Chemical energy: The molecules in gasoline (octane, the chemical formula shown) contain chemical energy. This energy is transformed into kinetic energy that allows a car to race on a racetrack.

Energy Changes in Chemical Reactions

Chemical reactions often produce changes in energy.

Learning Objectives

Describe the types of energy changes that can occur in chemical reactions

Key Takeaways

Key Points

- Chemical reactions often involve changes in energy due to the breaking and formation of bonds.
- Reactions in which energy is released are exothermic reactions, while those that take in heat energy are endothermic.

Key Terms

- endothermic: A description of a chemical reaction that absorbs heat energy from its surroundings.
- enthalpy: In thermodynamics, a measure of the heat content of a chemical or physical system. The change in enthalpy of a chemical reaction is symbolized as ΔH .
- exothermic: A description of a chemical reaction that releases heat energy to its surroundings.

Due to the absorption of energy when chemical bonds are broken, and the release of energy when chemical bonds are formed, chemical reactions almost always involve a change in energy between products and reactants. By the Law of Conservation of Energy, however, we know that the total energy of a system must remain unchanged, and that oftentimes a chemical reaction will absorb or release energy in the form of heat, light, or both. The energy change in a chemical reaction is due to the difference in the amounts of stored chemical energy between the products and the reactants.

This stored chemical energy, or heat content, of the system is known as its enthalpy.

Exothermic Reactions

Exothermic reactions release heat and light into their surroundings. For example, combustion reactions are usually exothermic. In exothermic reactions, the products have less enthalpy than the reactants, and as a result, an exothermic reaction is said to have a negative enthalpy of reaction. This means that the energy required to break the bonds in the reactants is less than the energy released when new bonds form in the products. Excess energy from the reaction is released as heat and light.

Chemical reaction: A thermite reaction, which produces molten iron.

Endothermic Reactions

Endothermic reactions, on the other hand, absorb heat and/or light from their surroundings. For example, decomposition reactions are usually endothermic. In endothermic reactions, the products have more enthalpy than the reactants. Thus, an endothermic reaction is said to have a positive enthalpy of reaction. This means that the energy required to break the bonds in the reactants is more than the energy released when new bonds form in the products; in other words, the reaction requires energy to proceed.

The decomposition of water into hydrogen and oxygen: When water is heated to over 2000 degrees Celsius, a small fraction will decompose into hydrogen and oxygen. Significant heat energy is needed for this reaction to proceed, so the reaction is endothermic.

Energy Use and the Environment

Present Sources of Energy

Present sources of energy include fossil fuels, various types of renewable energy, and nuclear power.

Learning Objectives

Identify the major present sources of energy

Key Takeaways

Key Points

- Fossil fuels are a finite source of energy that tend to release greenhouse gases as they are burned.
- Nuclear power is power that utilizes the intense energy stored in atoms.
- Renewable energy refers to energy from non-depletable sources such as solar, wind, and geothermal.

Key Terms

- fossil fuel: Any fuel derived from hydrocarbon deposits such as coal, petroleum, natural gas and, to some extent, peat. These fuels are irreplaceable and burning them generates the greenhouse gas carbon dioxide.
- renewable energy: Energy that can be replenished at the same rate as it is used.
- nuclear power: Power, especially electrical power, obtained using nuclear fission or nuclear fusion.

Present Levels of Earth's Energy Sources

Most of the world's energy resources are from the conversion of the sun's rays to other energy forms after being incident upon the planet. Some of that energy has been preserved as fossil energy

and some is directly or indirectly usable, such as via wind, hydro- or wave power.

The amount of energy the sun provides has been measured by satellite to be roughly 1368 watts per square meter, although this amount fluctuates by about 6.9% during the year due to the Earth's

varying distance from the sun. This value represents the total rate of solar power received by the planet, although only about half, 89 PW (PW = petawatt, which is equivalent to 10¹⁵ watts; a watt is equal to J/s), reaches the Earth's surface. The estimates for remaining non-renewable worldwide energy resources vary; the remaining fossil fuels total an estimated 0.4 YJ (1 YJ = yottajoule, or 10²⁴ J) and the the energy available from nuclear fuels such as uranium exceeds 2.5 YJ. Fossil fuels range from 0.6 to 3 YJ, if estimates of reserves of methane clathrates (methane gas trapped in ice deposits on the ocean floor and in Antarctic ice cores) are accurate, and if these methane clathrates

could become technically extractable. The total energy flux from the sun is 3.8 YJ/yr, which dwarfs all non-renewable resources.

Current Energy Consumption Trends

In 2008, 81% of our energy supply was from fossil fuels (33.5% oil, 26.8% coal, 20.8% gas, individually) while 12.9% was categorized as "other" (hydro, peat, solar, wind, geothermal power, biofuels, etc.), and 5.8% was nuclear. Oil was the most popular energy fuel, and oil and coal combined represented over 60% of the world energy supply in 2008. As of 2010, use of fossil fuels as an energy source comprised over 80% of total energy consumed.

Global energy use by source, 2010: Fossil fuels remain our primary source of energy. Renewable energy sources only comprised 16.7% of our energy in 2010.

Coal fueled the industrial revolution in the 18th and 19th centuries. However, with the advent of the automobile, airplanes, and the growing use of electricity among consumers, oil became the dominant fuel during the 20th century. The growth of oil as the largest fossil fuel was further enabled by steadily dropping prices from 1920 until 1973. However, after the oil shocks of 1973 and 1979, during which the price of oil increased from \$5 to \$45 per barrel, there was a shift away from this particular resource. Nowadays, coal, natural gas, and nuclear power are the most popular fuels for electricity generation.

Recent conservation measures have greatly increased energy efficiency. In the U.S., the average car has more than doubled the number of miles per gallon. Japan, which bore the brunt of the oil shocks, has also made spectacular improvements to its technology. Now Japan has the highest energy efficiency in the world.

Nuclear Power

As of December 2009, the world had 436 nuclear reactors. Since commercial nuclear energy began in the mid 1950's, 2008 was the first year that no new nuclear power plant was connected to the grid, although two were connected in 2009. Annual generation of nuclear power has been on a

slight downward trend since 2007, decreasing 1.8% in 2009 with nuclear power still meeting 13–14% of the world's electricity demand.

Nuclear power plant in Germany: Nuclear (fission) power stations, excluding the contribution from naval nuclear fission reactors, provided about 5.7% of the world's energy and 13% of the world's electricity in 2012.

Energy Consumption

Energy consumption is directly related to climate change.

Learning Objectives

Discuss energy consumption and its relationship with global warming

Key Takeaways

Key Points

- Fossil fuel -based energy use leads to the release of greenhouse gases.
- Since the Industrial Revolution, release of greenhouse gases into the atmosphere has risen exponentially.
- Impact of fossil fuel use is measured not just in terms of greenhouse gases produced, but also in terms of the environmental cost of extracting fossil fuels for use.

Key Terms

□ greenhouse gas: Any gas, such as carbon dioxide or CFCs, that contributes to the greenhouse effect when released into the atmosphere.

The environmental impact of the energy industry is diverse. Energy has been harnessed by humans for millennia. Initially, we used fire for light, heat, cooking, and safety. The use of fire can be traced back at least 1.9 million years. In recent years, however, there has been a trend towards the increased commercialization of various renewable energy sources.

Fossil Fuels and Climate Change

Wind turbines in Columbia: A wind turbine is a device that converts the natural kinetic energy from the wind into electrical power. Wind turbines provide a green source of alternative energy, as opposed to the burning of fossil fuels which contributes to climate change.

Consumption of fossil fuel resources has led to global warming and climate change. In most parts of the world, little effort is being made to slow these changes. If we reach (or have already reached) a maximum rate of petroleum extraction, then fossil fuels will no longer be a viable source of energy. However, if we explore viable alternative energy resources, we could reduce our impact on the environment.

Newly emerging technologies could lead to more efficient energy generation, which could temper environmental damage. The use of these new methods incorporate practices that are informed by new paradigms such as systems ecology and industrial ecology.

Global warming and climate change are generally accepted as being caused by anthropogenic (man-made) greenhouse gas emissions. The majority of greenhouse gas emissions are due to burning fossil fuels, while some is due to deforestation.

Global carbon emissions by energy source: Total carbon emissions has risen exponentially since the industrial revolution, from approximately 1 billion metric tons pre-1900 to 10 billion metric tons now.

The Future Costs of Fossil Fuels

There is a highly publicized denial of climate change, but the vast majority of scientists working in climatology accept that it is due to human activity. The IPCC report “Climate Change 2007: Climate Change Impacts, Adaptation, and Vulnerability,” predicts that climate change will cause shortages of food and water, as well as increase flooding that will affect billions of people. Those living in poverty will be most affected.

Another measurement of greenhouse gas-related issues and other externality comparisons between

energy sources can be found in the ExternE project by the Paul Scherrer Institute and the University of Stuttgart. According to this study, hydroelectricity produces the lowest CO₂ emissions, wind produces the second lowest CO₂ emissions, nuclear energy produces the third lowest, and solar photovoltaic produces the fourth lowest.

Similarly, the same research study, known as ExternE, or “Externalities of Energy,” which was undertaken over the period from 1995 to 2005, found that the cost of producing electricity from coal or oil would double over its present value. The cost of electricity production from gas would increase by 30% if external costs such as damage to the environment and to human health, from the airborne particulate matter, nitrogen oxides, chromium VI, and arsenic emissions produced by these sources, were taken into account. It was estimated in the study that these external downstream

fossil fuel costs amount up to 1%-2% of the EU’s entire gross domestic product (GDP); this was before the external cost of global warming from these sources was even included. The study also found that the environmental and health costs of nuclear power, per unit of energy delivered, was €0.0019/kWh, which was found to be lower than that of many renewable sources including biomass and photovoltaic solar panels. It was 30 times lower than coal at €0.06/kWh, or 6 cents/kWh. The energy source with the lowest associated external environmental and health costs was wind power at €0.0009/kWh.

Environmental Problems Associated with Fossil Fuel Use

One of the major environmental problems associated with fossil fuel use is global warming.

Learning Objectives

Relate fossil fuels to an increase in the earth’s temperature

Key Takeaways

Key Points

- Human energy use has led to an increase in the production of greenhouse gases.
- Humans produce greenhouse gases faster than they can be taken up by carbon users, such as plants.
- This increase in greenhouse gases has led to a continued, irreversible rise in planetary

temperature.

Key Terms

- anthropogenic: Having its origin in the influence of human activity on nature.
- greenhouse gas: Any gas, such as carbon dioxide or CFCs, that contributes to the greenhouse effect when released into the atmosphere.
- global warming: A sustained increase in the average temperature of the earth, sufficient to cause climate change.

One of the biggest environmental problems associated with fossil fuel use is global warming, which is the rise in the average temperature of Earth's atmosphere and oceans since the late 19th century. Since the early 20th century, Earth's mean surface temperature has increased by about 0.8

°C (1.4 °F), with about two-thirds of the increase occurring since 1980. This warning of the climate system is unequivocal, and scientists are now more than 90% certain that it is primarily caused by increasing concentrations of greenhouse gases produced by human activities such as deforestation and the burning of fossil fuels. These findings are recognized by the national science academies of all major industrialized nations.

Global temperature, 1880-2010: Global temperature has risen steadily since the industrial revolution. The graph shows an increase of about 1 degree Celsius since 1910.

Current Projections of Global Warming

Climate model projections were summarized in the 2007 Fourth Assessment Report (AR4) by the Intergovernmental Panel on Climate Change (IPCC). They indicated that during the 21st century, the global surface temperature is likely to rise another 1.1 to 2.9 °C (2 to 5.2 °F) for their lowest emissions scenario, and 2.4 to 6.4 °C (4.3 to 11.5 °F) for their highest. The ranges of these estimates arise from the use of models with differing sensitivity to greenhouse gas concentrations.

According to AR4, warming and related changes will vary from region to region around the globe. The effects of an increase in global temperature include a rise in sea levels, a change in the amount and pattern of precipitation, and a probable expansion of subtropical deserts. Warming is expected to be strongest in the Arctic and would be associated with the continuing retreat of glaciers, permafrost, and sea ice. Other likely effects of global warming include a more frequent occurrence of extreme weather events such as heat waves, droughts, and heavy rainfall. Ocean acidification will worsen, and there will also be species extinctions due to shifting temperature regimes. Effects significant to humans include the threat to food security from decreasing crop yields, as well as the loss of habitat from flooding.

Arctic sea ice loss: This time series, based on satellite data, shows the annual Arctic sea ice minimum since 1979. It shows a steady decrease over time; the September 2010 extent was the third lowest in the satellite record.

Current Efforts Aimed at Reducing Climate Change

Proposed policy responses to global warming include mitigation by emissions reduction, adaptation to its effects, and possible future geoengineering. Most countries are parties to the

United Nations Framework Convention on Climate Change (UNFCCC), whose ultimate objective is to prevent dangerous anthropogenic climate change. Parties to the UNFCCC have adopted a range of policies designed to reduce greenhouse gas emissions and to assist in adaptation to global warming. This group has agreed that deep cuts in emissions are required, and that future global warming should be limited to below 2.0 °C (3.6 °F) relative to the pre-industrial level. Reports published in 2011 by the United Nations Environment Programme and the International Energy Agency suggest that efforts as of the early 21st century to reduce emissions may be inadequate to meet the UNFCCC's 2.0 °C target.

New Energy Sources

Alternative and renewable energy sources can reduce the environmental impact of energy production and consumption.

Learning Objectives

Discuss new energy sources and their effects

Key Takeaways

Key Points

- Alternative energy is a form of renewable energy.
- Renewable energy capacity is increasing as more countries introduce limits on carbon emissions.
- As technology continues to improve, it is projected that the world will continue to shift to these forms of energy production.

Key Terms

- renewable energy: Energy that can be replenished at the same rate as it is used.
- alternative energy: Energy derived from any renewable source; i.e., energy not from fossil fuels or nuclear fission.
- biomass: An alternative energy source consisting of wood, corn, and some types of garbage, which can be burned for fuel.

Renewable Energy

Renewable energy is energy that comes from natural resources, such as sunlight, wind, rain, tides, waves, and geothermal heat, which are all naturally replenished. About 16% of global final energy consumption comes from renewable sources, with 10% coming from traditional biomass, which is the burning of natural materials such as wood, corn, and some types of garbage to provide energy

mainly used for heating, and 3.4% from hydroelectricity. New renewables (small hydro, modern biomass, wind, solar, geothermal, and biofuels) account for another 3% and are growing very rapidly. The share of renewables in electricity generation is around 19%, with 16% of global electricity coming from hydroelectricity, and 3% from new renewable sources.

Global energy capacity expansion by renewable source: Total renewable power capacity has been increasing over the past several years, from roughly 100 GW in 2005 to nearly 400 GW in

2007.

Various Alternative Energy Sources Worldwide

The use of wind power is increasing at an annual rate of 20%. At the end of 2011, a worldwide installed capacity of 238,000 megawatts (MW) was in use widely throughout Europe, Asia, and the United States. Since 2004, photovoltaics (solar cells) surpassed wind as the fastest growing energy source, and since 2007, the use of solar power has more than doubled every two years. At the end of 2011, the photovoltaic (PV) capacity worldwide was 67,000 MW. PV power stations are becoming increasingly popular in Germany and Italy. In addition, solar thermal power stations, which are power plants that generate electricity from the heat of the sun's rays, operate in the USA and Spain. The largest of these is the 354 MW SEGS power plant located in the Mojave Desert. The world's largest geothermal power installation is the Geysers complex in California, with a rated capacity of 750 MW. Brazil has one of the largest renewable energy programs in the world, involving production of ethanol fuel from sugarcane. Ethanol now provides 18% of the country's automotive fuel. Ethanol fuel is also widely available in the USA.

Sugarcane being harvested for ethanol production: Ethanol is a quasi-renewable energy source. This is because, while the energy is partially generated by non-depletable sunlight, the harvesting process requires vast amounts of energy that typically comes from non-renewable sources.

While many renewable energy projects are large-scale, renewable technologies can also be suited to rural and remote areas, where energy is often crucial in human development. As of 2011, small solar PV systems provide electricity to a few million households, and micro-hydroelectricity plants configured into mini-grids serve many more. Over 44 million households use biogas (mixtures of gases produced by the breakdown of organic matter) made in household-scale digesters for lighting and/or cooking, and more than 166 million households rely on a new generation of more efficient biomass cookstoves.

The Future of Alternative Energy

United Nations' Secretary-General Ban Ki-moon has said that renewable energy has the ability to lift the poorest nations to new levels of prosperity. Carbon neutral and negative fuels can be stored and transported through existing natural gas pipelines. Using the existing transportation infrastructure, thereby displacing fossil fuels, will reduce greenhouse gas emissions.

Climate change concerns, coupled with high oil prices, peak oil, and increasing government support, are driving more and more renewable energy legislation, incentives and commercialization. New government spending, regulation and policies are helping the industry weather the global financial crisis better than many other sectors. According to a 2011 projection by the International Energy Agency, solar power generators may produce most of the world's electricity within 50 years, dramatically reducing the emissions of greenhouse gases that harm the environment.

Enthalpy

Internal Energy and Enthalpy

The enthalpy of reaction measures the heat released/absorbed by a reaction that occurs at constant pressure.

Learning Objectives

Review enthalpy of reaction

Key Takeaways

Key Points

- At constant volume, the heat of reaction is equal to the change in the internal energy of the system.
- At constant pressure, the heat of reaction is equal to the enthalpy change of the system.
- Most chemical reactions occur at constant pressure, so enthalpy is more often used to measure heats of reaction than internal energy.

Key Terms

- enthalpy: In thermodynamics, a measure of the heat content of a chemical or physical system.
- internal energy: A property characteristic of the state of a thermodynamic system, the change in which is equal to the heat absorbed minus the work done by the system.
- first law of thermodynamics: Heat and work are forms of energy transfer; the internal energy of a closed system changes as heat and work are transferred into or out of it.

In thermodynamics, work (W) is defined as the process of an energy transfer from one system to another. The first law of thermodynamics states that the energy of a closed system is equal to the amount of heat supplied to the system minus the amount of work done by the system on its surroundings. The amount of energy for a closed system is written as follows:

$$\Delta U = Q - W$$

In this equation, U is the total energy of the system, Q is heat, and W is work. In chemical systems, the most common type of work is pressure-volume (PV) work, in which the volume of a gas changes. Substituting this in for work in the above equation, we can define the change in internal energy for a chemical system:

$$\Delta U = Q - P\Delta V$$

Internal Energy Change at Constant Volume

Let's examine the internal energy change, ΔU , at constant volume. At constant volume, $\Delta V = 0$, the equation for the change in internal energy reduces to the following:

$$\Delta U = Q_V$$

The subscript V is added to Q to indicate that this is the heat transfer associated with a chemical process at constant volume. This internal energy is often very difficult to calculate in real life settings, though, because chemists tend to run their reactions in open flasks and beakers that allow gases to escape to the atmosphere. Therefore, volume is not held constant, and calculating ΔU becomes problematic. To correct for this, we introduce the concept of enthalpy, which is much

more commonly used by chemists.

Standard Enthalpy of Reaction

The enthalpy of reaction is defined as the internal energy of the reaction system, plus the product of pressure and volume. It is given by:

$$H = U + PV \quad \Delta H = \Delta U + P\Delta V$$

By adding the PV term, it becomes possible to measure a change in energy within a chemical system, even when that system does work on its surroundings. Most often, we are interested in the change in enthalpy of a given reaction, which can be expressed as follows:

$$\Delta H = \Delta U + P\Delta V \quad \Delta H = \Delta U + P\Delta V$$

When you run a chemical reaction in a laboratory, the reaction occurs at constant pressure, because

the atmospheric pressure around us is relatively constant. We will examine the change in enthalpy for a reaction at constant pressure, in order to see why enthalpy is such a useful concept for chemists.

Enthalpy of Reaction at Constant Pressure

Let's look once again at the change in enthalpy for a given chemical process. It is given as follows:

$$\Delta H = \Delta U + P\Delta V \quad \Delta H = \Delta U + P\Delta V$$

However, we also know that:

$$\Delta U = Q - W = Q - P\Delta V \quad \Delta U = Q - W = Q - P\Delta V$$

Substituting to combine these two equations, we have:

$$\Delta H = Q - P\Delta V + P\Delta V = Q \quad \Delta H = Q - P\Delta V + P\Delta V = Q$$

Thus, at constant pressure, the change in enthalpy is simply equal to the heat released/absorbed by

the reaction. Due to this relation, the change in enthalpy is often referred to simply as the "heat of reaction."

Enthalpy: An explanation of why enthalpy can be viewed as "heat content" in a constant pressure system.

Exothermic and Endothermic Processes

Endothermic reactions absorb energy from the environment, while exothermic reactions release energy to the environment.

Learning Objectives

Distinguish between endothermic and exothermic reactions

Key Takeaways

Key Points

- All chemical reactions involve the transfer of energy.
- Endothermic processes require an input of energy to proceed and are signified by a positive change in enthalpy.

□ Exothermic processes release energy upon completion, and are signified by a negative change in enthalpy.

Key Terms

□ exothermic: Of a chemical reaction that releases energy in the form of heat.

□ enthalpy: In thermodynamics, a measure of the heat content of a chemical or physical system.

□ endothermic: Of a chemical reaction that absorbs heat energy from its surroundings.

All chemical processes are accompanied by energy changes. When a reaction proceeds, it either releases energy to, or absorbs energy from, its surroundings. In thermodynamics, these two types of reactions are classified as exothermic or endothermic, respectively. An easy way to remember the difference between these two reaction types is by their prefixes: endo- means to draw in, and exo- means to give off. We will explore these concepts in more detail after introducing the concept of enthalpy.

Enthalpy

Enthalpy (signified as H) is a measure of the total energy of a system and often expresses and simplifies energy transfer between systems. Since the total enthalpy of a system cannot be measured directly, we most often refer to the change in enthalpy for a particular chemical reaction. At constant pressure, the change in enthalpy is equal to the heat given off, or the heat absorbed, in a given chemical reaction:

$$\Delta H = q_{rxn} \quad \Delta H = q_{rxn}$$

Due to this relation, the change in enthalpy, ΔH , is often referred to as the “heat of reaction.”

Exothermic Reactions

Exothermic reactions are reactions or processes that release energy, usually in the form of heat or light. In an exothermic reaction, energy is released because the total energy of the products is less than the total energy of the reactants. For this reason, the change in enthalpy, ΔH , for an exothermic reaction will always be negative. In the presence of water, a strong acid will dissociate quickly and release heat, so it is an exothermic reaction.

Exothermic reaction: In an exothermic reaction, the total energy of the products is less than the total energy of the reactants. Therefore, the change in enthalpy is negative, and heat is released to the surroundings.

Endothermic Reactions

Endothermic reactions are reactions that require external energy, usually in the form of heat, for the reaction to proceed. Since endothermic reactions draw in heat from their surroundings, they tend to cause their environments to cool down. They are also generally non-spontaneous, since endothermic reactions yield products that are higher in energy than the reactants. As such, the change in enthalpy for an endothermic reaction is always positive. In order to melt the ice cube, heat is required, so the process is endothermic.

Endothermic reaction: In an endothermic reaction, the products are higher in energy than the

reactants. Therefore, the change in enthalpy is positive, and heat is absorbed from the surroundings by the reaction.

Whether a reaction is endothermic or exothermic depends on the direction that it is going; some reactions are reversible, and when you revert the products back to reactants, the change in enthalpy is opposite.

Endothermic and exothermic reactions: Paul Andersen explains how heat can be absorbed in endothermic or released in exothermic reactions. An energy diagram can be used to show energy movements in these reactions and temperature can be used to measure them macroscopically.

Thermochemical Equations

Thermochemical equations are chemical equations which include the enthalpy change of the reaction, ΔH_{rxn} .

Learning Objectives

Give examples of thermochemical equations

Key Takeaways

Key Points

- If ΔH is negative, the reaction is exothermic; if ΔH is positive, the reaction is endothermic.
- Thermochemical equations have the general form: $A + B \rightarrow C, \Delta H = (\pm n)$
- The value of enthalpy is dependent on the reaction conditions, as well as the concentration of reactants and products.

Key Terms

- thermochemical equation: A special equation type, denoting the overall change in energy.
- enthalpy: In thermodynamics, a measure of the heat content of a chemical or physical system.
- Hess's law: States that the enthalpy change for a reaction is the same whether it occurs in one step or in a series of steps.

Enthalpy (H) is a measure of the energy in a system, and the change in enthalpy is denoted by ΔH . Since enthalpy is a state function, the value of ΔH is independent of the path taken by the reactions to reach the products. Values of ΔH can be determined experimentally under standard conditions.

A thermochemical equation is a balanced stoichiometric chemical equation which includes the enthalpy change. The equations take the form: $A + B \rightarrow C, \Delta H = (\pm n)$

Thermochemical Equations for Endothermic Reactions

The sign of the ΔH value indicates whether or not the system is endothermic or exothermic. In an endothermic system, the ΔH value is positive, so the reaction absorbs heat into the system. The equation takes the form:

heat + A + B → C, ΔH = + heat + A + B → C, ΔH = +

Notice that in an endothermic reaction like the one depicted above, we can think of heat as being a reactant, just like A and B.

Thermochemical Equations for Exothermic Reactions

In an exothermic system, the ΔH value is negative, so heat is given off by the reaction. The equation takes the form:

A + B → C + heat, ΔH = - A + B → C + heat, ΔH = -

Notice that here, we can think of heat as being a product in the reaction.

ΔH is dependent on both the phase (solid, liquid, or gas) as well as the molar ratios of the reactants and products. Therefore, all thermochemical equations must be stoichiometrically balanced. This becomes important once we begin working with Hess's law.

Thermochemical equations: Thermochemical equations can describe endothermic or exothermic reactions.

Processing math: 47%

Standard Enthalpy of Formation and Reaction

Standard States and Standard Enthalpy Changes

The standard enthalpy of formation refers to the enthalpy change when one mole of a compound is formed from its elements.

Learning Objectives

Calculate the standard enthalpy of formation

Key Takeaways

Key Points

- The standard state of a material is a reference point for the material's thermodynamic state properties such as enthalpy, entropy, Gibbs free energy, etc. It is used to calculate the material's properties under different conditions and is denoted as H^o_f.
- The standard state for a gas is the hypothetical state at 1 bar; for liquids and solids, the pure substance at 1 bar; for elements, the most stable allotrope of the element; and for a substance in solution (solute), concentration at 1 M and 1 bar.
- The standard enthalpy of formation is the change in enthalpy that accompanies the formation of one mole of the compound from its elements. The standard enthalpy of reaction occurs in a system when one mole of matter is transformed by a chemical reaction.

Key Terms

- standard state: In chemistry, a reference point used to calculate a material's (pure substance, mixture, or solution) properties under different conditions.
- standard enthalpy of formation: The change in enthalpy that accompanies the formation of one mole of a compound from its elements, with all substances in their standard states; also called "standard heat of formation."
- enthalpy of solution: The heat association with dissolving a particular solute in a particular

solvent.

Standard States

In chemistry, the standard state of a material, be it a pure substance, mixture, or solution, is a reference point used to calculate its properties under different conditions. In principle, the choice of standard state is arbitrary, although the International Union of Pure and Applied Chemistry (IUPAC) recommends a conventional set of standard states for general use. A standard pressure of 1 bar (101.3 kilopascals) has been accepted.

Strictly speaking, temperature is not part of the definition of a standard state; the standard state of a gas is conventionally chosen to be 1 bar for an ideal gas, regardless of the temperature. However, most tables of thermodynamic quantities are compiled at specific temperatures, most commonly 298.15 K (exactly 25°C) or, somewhat less commonly, 273.15 K (exactly 0°C).

Standard states for atomic elements are given in terms of the most stable allotrope for each element.

For example, white tin and graphite are the most stable allotropes of tin and carbon, respectively. Therefore, they are used as standard states or reference points for calculations of different thermodynamic properties of these elements.

Tin: White tin (on the left) is the most stable allotrope of tin, and is used as its standard state for thermodynamic calculations.

The standard state should not be confused with standard temperature and pressure (STP) for gases,

or with the standard solutions used in analytical chemistry. Standard states are often indicated in textbooks by a circle with a horizontal bar $\text{H}\ominus\text{f}$.

Graphite: Graphite is the most stable state of carbon and is used in thermochemistry to define the heat of formation of carbon compounds.

Standard Enthalpy of Formation

The standard enthalpy of formation, or standard heat of formation, of a compound is the change in enthalpy that accompanies the formation of one mole of the compound from its elements in their standard states. For example, the standard enthalpy of formation for carbon dioxide would be the change in enthalpy for the following reaction:



Note that standard enthalpies of formation are always given in units of kJ/mol of the compound formed.

Standard Enthalpy of Reaction

The standard enthalpy of reaction is the enthalpy change that occurs in a system when a chemical reaction transforms one mole of matter under standard conditions.

Learning Objectives

Demonstrate how to directly calculate the standard enthalpy of reaction

Key Takeaways

Key Points

□ The standard enthalpy of reaction, ΔH_{rxn} , can be calculated by summing the standard enthalpies of formation of the reactants and subtracting the value from the sum of the standard enthalpies of formation of the products.

□ The following equation can be used to calculate the standard enthalpy of reaction:

$$\Delta H_{\text{rxn}} = \sum \Delta H_{\text{f}}^{\circ} \{\text{products}\} - \sum \Delta H_{\text{f}}^{\circ} \{\text{reactants}\}.$$

□ The enthalpy of reaction is calculated under standard conditions (STP).

Key Terms

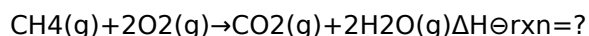
□ standard enthalpy of reaction: The enthalpy change that occurs in a system when one mole of matter is transformed by a chemical reaction under standard conditions.

The standard enthalpy of reaction, ΔH_{rxn} , is the change in enthalpy for a given reaction calculated from the standard enthalpies of formation for all reactants and products. The change in enthalpy does not depend upon the particular pathway of a reaction, but only upon the overall energy level of the products and reactants; enthalpy is a state function, and as such, it is additive. In order to calculate the standard enthalpy of a reaction, we can sum up the standard enthalpies of formation of the reactants and subtract this from the sum of the standard enthalpies of formation of the products. Stated mathematically, this gives us:

$$\Delta H_{\text{rxn}} = \sum \Delta H_{\text{f}}^{\circ} \{\text{products}\} - \sum \Delta H_{\text{f}}^{\circ} \{\text{reactants}\}$$

Calculating the Standard Enthalpy of Reaction

Calculate the standard enthalpy of reaction for the combustion of methane:



In order to calculate the standard enthalpy of reaction, we need to look up the standard enthalpies of formation for each of the reactants and products involved in the reaction. These are typically found in an appendix or in various tables online. For this reaction, the data we need is:

$$\Delta H_{\text{f}}^{\circ} \{\text{CH}_4(\text{g})\} = -75 \text{ kJ/mol}$$

$$\Delta H_{\text{f}}^{\circ} \{\text{O}_2(\text{g})\} = 0 \text{ kJ/mol}$$

$$\Delta H_{\text{f}}^{\circ} \{\text{CO}_2(\text{g})\} = -394 \text{ kJ/mol}$$

$$\Delta H_{\text{f}}^{\circ} \{\text{H}_2\text{O}(\text{g})\} = -284 \text{ kJ/mol}$$

Note that because it exists in its standard state, the standard enthalpy of formation for oxygen gas is 0 kJ/mol. Next, we sum up our standard enthalpies of formation. Keep in mind that because the units are in kJ/mol, we need to multiply by the stoichiometric coefficients in the balanced reaction equation.

$$\sum \Delta H_{\text{f}}^{\circ} \{\text{products}\} = \Delta H_{\text{f}}^{\circ} \{\text{CO}_2(\text{g})\} + 2 \Delta H_{\text{f}}^{\circ} \{\text{H}_2\text{O}(\text{g})\} = (1)(-394) + (2)(-284) = -962 \text{ kJ/mol}$$

$$\sum \Delta H_{\text{f}}^{\circ} \{\text{reactants}\} = \Delta H_{\text{f}}^{\circ} \{\text{CH}_4(\text{g})\} + 2 \Delta H_{\text{f}}^{\circ} \{\text{O}_2(\text{g})\} = (1)(-75) + (2)(0) = -75 \text{ kJ/mol}$$

Now, we can find the standard enthalpy of reaction:

$$\Delta H^{\circ}_{\text{rxn}} = \sum \Delta H^{\circ}_{\text{f}}(\text{products}) - \sum \Delta H^{\circ}_{\text{f}}(\text{reactants}) = (-962) - (-75) = -887 \text{ kJ/mol}$$

As we would expect, the standard enthalpy for this combustion reaction is strongly exothermic.

A calculation of standard enthalpy of reaction ($\Delta H^{\circ}_{\text{rxn}}$) from standard heats of formation ($\Delta H^{\circ}_{\text{f}}$): A standard enthalpy of reaction ($\Delta H^{\circ}_{\text{rxn}}$) problem, involving ethylene and oxygen as reactants to yield carbon dioxide and gaseous water, is shown.

Hess's Law

Hess's Law sums the changes in enthalpy for a series of intermediate reaction steps to find the overall change in enthalpy for a reaction.

Learning Objectives

Employ Hess's law to determine $\Delta H^{\circ}_{\text{rxn}}$

Key Takeaways

Key Points

- Hess's law states that the standard reaction enthalpy is the sum of the standard enthalpies of the intermediate reactions into which the overall reaction can be divided, while each occurs at the same temperature.
- Enthalpy change for a reaction is independent of the number of ways a product can be obtained, if the initial and final conditions are the same.
- Negative enthalpy change for a reaction indicates exothermic process, while positive enthalpy change corresponds to endothermic process.

Key Terms

- Hess's law: States that, if an overall reaction takes place in several steps, its standard reaction enthalpy is the sum of the standard enthalpies of the intermediate reactions, at the same temperature.
- law of conservation of energy: States that the total amount of energy in any isolated system remains constant; energy cannot be created or destroyed, although it can change forms.

Derivation of Hess's Law

Hess's law is a relationship in physical chemistry named after Germain Hess, a Swiss-born Russian chemist and physician. This law states that if a reaction takes place in several steps, then the standard reaction enthalpy for the overall reaction is equal to the sum of the standard enthalpies of the intermediate reaction steps, assuming each step takes place at the same temperature.

Hess's law derives directly from the law of conservation of energy, as well as its expression in the first law of thermodynamics. Since enthalpy is a state function, the change in enthalpy between products and reactants in a chemical system is independent of the pathway taken from the initial to the final state of the system. Hess's law can be used to determine the overall energy required for a chemical reaction, especially when the reaction can be divided into several intermediate steps

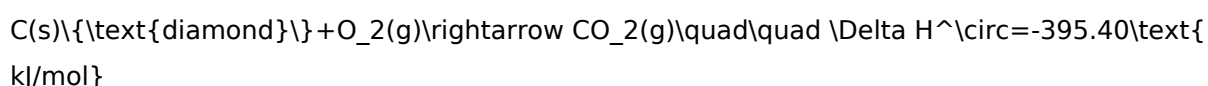
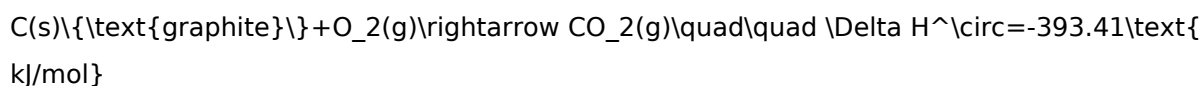
that are individually easier to characterize. Negative enthalpy change for a reaction indicates exothermic process, while positive enthalpy change corresponds to endothermic process.

Graphical representation of Hess's law: The net reaction here is A being converted into D, and the change in enthalpy for that reaction is ΔH . However, we can see that the net reaction is a result of A being converted into B, which is then converted into C, which is finally converted into D. By Hess's law, the net change in enthalpy of the overall reaction is equal to the sum of the changes in enthalpy for each intermediate transformation: $\Delta H = \Delta H_1 + \Delta H_2 + \Delta H_3$.

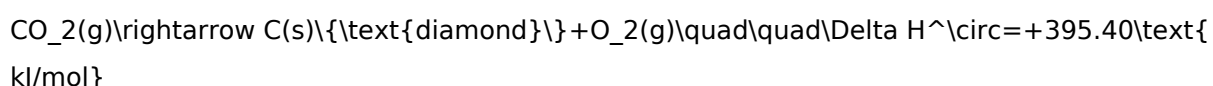
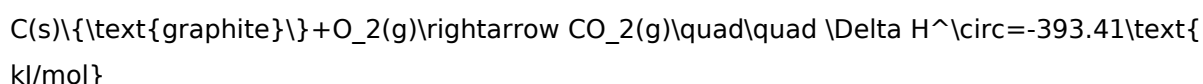
Calculating Standard Enthalpies of Reaction Using Hess's Law



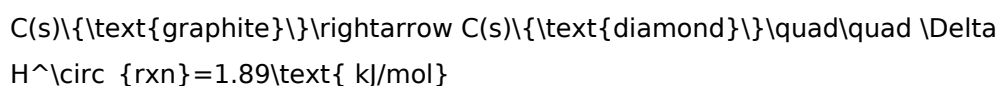
Turning graphite into diamond requires extremely high temperatures and pressures, and therefore is impractical in a laboratory setting. The change in enthalpy for this reaction cannot be determined experimentally. However, because we know the standard enthalpy change for the oxidation for these two substances, it is possible to calculate the enthalpy change for this reaction using Hess's law. Our intermediate steps are as follows:



In order to get these intermediate reactions to add to our net overall reaction, we need to reverse the second step. Keep in mind that when reversing reactions using Hess's law, the sign of ΔH will change. Sometimes, you will need to multiply a given reaction intermediate through by an integer. In such cases, you need always multiply your ΔH value by that same integer. Restating the first equation and flipping the second equation, we have:



Adding these equations together, carbon dioxides and oxygens cancel, leaving us only with our net equation. By Hess's law, we can sum the ΔH values for these intermediate reactions to get our final value, $\Delta H^\circ_{\text{rxn}}$.



Hess's law lesson: This lesson uses two methods to find the heat of reaction for a given reaction. First it looks at combining reactions according to Hess's law and their heats of reaction, and then it discusses using standard heats of formation of the reactants and products to find the overall heat of reaction.

Heat of Solution

Heat of solution refers to the change in enthalpy when a solute is dissolved into a solvent.

Learning Objectives

Define heat of solution

Key Takeaways

Key Points

□ Enthalpy of solution, or heat of solution, is expressed in kJ/mol, and it is the amount of heat energy that is released or absorbed when a solution is formed.

□ There are three steps in solvation: the breaking of bonds between solute molecules, the breaking of intermolecular attractions between solvent molecules, and the formation of new solute-solvent attractive bonds. Energy is absorbed during the first two steps, and it is released during the last step.

□ Depending on the relative amounts of energy required to break bonds initially, as well as how much is released upon solute-solvent bond formation, the overall heat of solution can either be endothermic or exothermic.

Key Terms

□ heat of solution: The enthalpy change associated with the dissolution of a substance in a solvent at constant pressure, resulting in infinite dilution.

□ solvation: The process of attraction and association of molecules of a solvent with molecules or ions of a solute; also called dissolution.

The heat of solution, also referred to the enthalpy of solution or enthalpy of dissolution, is the enthalpy change associated with the dissolution of a solute in a solvent at constant pressure, resulting in infinite dilution. The heat of solution, like all enthalpy changes, is expressed in kJ/mol for a reaction taking place at standard conditions (298.15 K and 1 bar).

Three-Step Process of Dissolution

The heat of solution can be regarded as the sum of the enthalpy changes of three intermediate steps:

1. The breaking of bonds within the solute, such as the electrostatic attraction between two ions (endothermic)
2. The breaking of intermolecular attractive forces within the solvent, such as hydrogen bonds (endothermic)
3. The formation of new attractive solute-solvent bonds in solution (exothermic)

The value of the overall heat of solution, $\Delta H^{\circ}_{\text{sol}}$, is the sum of these individual steps. Depending on the relative signs and magnitudes of each step, the overall heat of solution can be either positive or negative, and therefore either endothermic or exothermic. This depends entirely on if more energy was used to break the solute-solute and solvent-solvent bonds, or if more energy was released when solute-solvent bonds were formed.

If more energy is released in making bonds than is used in breaking bonds, the overall process is exothermic, and ΔH_{sol} is negative. If more energy is used in breaking bonds than is released upon solute-solvent bond formation, then the overall process is endothermic, and ΔH_{sol} is positive.

Examples

□ Dissolution of sodium chloride (table salt) in water is endothermic. This is because the amount of energy used to break apart the hydrogen bonding interactions between water molecules, as well as the energy used to break apart the electrostatic attractions between sodium and chloride ions, is greater than the amount of energy released when new solute-solvent attractions are formed between water molecules and aqueous ions in solution.

□ Dissolving potassium hydroxide is exothermic. This is because more energy is released upon formation of solute-solvent bonds than was required to break apart the hydrogen bonds in water, as well as the ionic bonds in KOH.

Dissolution of NaCl in water: Dissolution of sodium chloride in water is endothermic. Solute-solvent attractive bond formation (the exothermic step in the process of solvation) is indicated by dashed lines.

Introduction to Thermodynamics

The First Law of Thermodynamics

The first law of thermodynamics states that energy can be transferred or transformed, but cannot be created or destroyed.

Learning Objectives

Describe the first law of thermodynamics

Key Takeaways

Key Points

□ According to the first law of thermodynamics, the total amount of energy in the universe is constant.

□ Energy can be transferred from place to place or transformed into different forms, but it cannot be created or destroyed.

□ Living organisms have evolved to obtain energy from their surroundings in forms that they can transfer or transform into usable energy to do work.

Key Terms

□ first law of thermodynamics: A version of the law of conservation of energy, specialized for thermodynamical systems, that states that the energy of an isolated system is constant and can neither be created nor destroyed.

□ work: A measure of energy expended by moving an object, usually considered to be force times distance. No work is done if the object does not move.

Thermodynamics is the study of heat energy and other types of energy, such as work, and the various ways energy is transferred within chemical systems. “Thermo-” refers to heat, while “dynamics” refers to motion.

The First Law of Thermodynamics

The first law of thermodynamics deals with the total amount of energy in the universe. The law states that this total amount of energy is constant. In other words, there has always been, and always

will be, exactly the same amount of energy in the universe.

Energy exists in many different forms. According to the first law of thermodynamics, energy can be transferred from place to place or changed between different forms, but it cannot be created or destroyed. The transfers and transformations of energy take place around us all the time. For instance, light bulbs transform electrical energy into light energy, and gas stoves transform chemical energy from natural gas into heat energy. Plants perform one of the most biologically useful transformations of energy on Earth: they convert the energy of sunlight into the chemical energy stored within organic molecules.

The first law of thermodynamics: Shown are two examples of energy being transferred from one system to another and transformed from one form to another. Humans can convert the chemical energy in food, like this ice cream cone, into kinetic energy by riding a bicycle. Plants can convert electromagnetic radiation (light energy) from the sun into chemical energy.

The System and Surroundings

Thermodynamics often divides the universe into two categories: the system and its surroundings. In chemistry, the system almost always refers to a given chemical reaction and the container in which it takes place. The first law of thermodynamics tells us that energy can neither be created nor destroyed, so we know that the energy that is absorbed in an endothermic chemical reaction must have been lost from the surroundings. Conversely, in an exothermic reaction, the heat that is released in the reaction is given off and absorbed by the surroundings. Stated mathematically, we have:

$$\Delta E = \Delta E_{\text{sys}} + \Delta E_{\text{surr}} = 0 \quad \Delta E = \Delta E_{\text{sys}} + \Delta E_{\text{surr}} = 0$$

The system and surroundings: A basic diagram showing the fundamental distinction between the system and its surroundings in thermodynamics.

Heat and Work

We know that chemical systems can either absorb heat from their surroundings, if the reaction is endothermic, or release heat to their surroundings, if the reaction is exothermic. However, chemical reactions are often used to do work instead of just exchanging heat. For instance, when rocket fuel burns and causes a space shuttle to lift off from the ground, the chemical reaction, by propelling the rocket, is doing work by applying a force over a distance.

If you've ever witnessed a video of a space shuttle lifting off, the chemical reaction that occurs also releases tremendous amounts of heat and light. Another useful form of the first law of thermodynamics relates heat and work for the change in energy of the internal system:

$$\Delta E_{\text{sys}} = Q + W \quad \Delta E_{\text{sys}} = Q + W$$

While this formulation is more commonly used in physics, it is still important to know for chemistry.

Rocket launch: The powerful chemical reaction propelling the rocket lets off tremendous heat to the surroundings and does work on the surroundings (the rocket) as well.

Heat and Work

Both heat and work refer to processes by which energy is transferred to or from a substance.

Learning Objectives

Define heat and work

Key Takeaways

Key Points

- When energy is exchanged between thermodynamic systems by thermal interaction, the transfer of energy is called heat.
- Work is the transfer of energy by any process other than heat.
- Heat and work are related: work can be completely converted into heat, but the reverse is not true: heat cannot be completely converted to work.

Key Terms

- heat: The energy transferred from one system to another by thermal interaction.
- work: The transfer of energy by any process other than heat.
- thermochemistry: The study of the energy and heat associated with chemical reactions and/or physical transformations.

Definition of Heat

When energy is exchanged between thermodynamic systems by thermal interaction, the transfer of energy is called heat. The units of heat are therefore the units of energy, or joules (J). Heat is transferred by conduction, convection, and/or radiation.

Heat is transfer by conduction occurs when an object with high thermal energy comes into contact with an object with low thermal energy. Heat transfer by convection occurs through a medium. For example, when heat transfers from the hot water at the bottom of the pot to the cooler water at the top of the pot. Lastly, heat can also be transferred by radiation; a hot object can convey heat to anything in its surroundings via electromagnetic radiation.

When a high temperature body is brought into contact with a low temperature body, the temperatures equilibrate: there is heat flow from higher to lower temperature, like water flowing downhill, until the temperatures of the bodies are equivalent. The high temperature body loses thermal energy, and the low temperature body acquires this same amount of thermal energy. The system is then said to be at thermal equilibrium.

An illustration of thermal equilibrium: The can of cola and ice cube start at different temperatures. When they come into contact, heat is transferred from the cola can to the ice cube until both bodies reach thermal equilibrium.

Definition of Work

Work is the transfer of energy by any process other than heat. Like heat, the unit measurement for work is joules (J). There are many forms of work, including but not limited to mechanical, electrical, and gravitational work. For our purposes, we are concerned with P-V work, which is the work done in an enclosed chemical system. In this type of system, work is defined as the change

in the volume (V) in liters within the system multiplied by a pressure (P). Assuming the system is

at constant pressure, this equates to the following:

$$W = P\Delta V = P\Delta V$$

Most often, we are interested in the work done by expanding gases. Assuming the gases are ideal, we can apply the ideal gas law to the above equation to get the following:

$$W = P\Delta V = nR\Delta T \quad W = P\Delta V = nR\Delta T$$

Relationship Between Heat and Work

Heat and work are related. Work can be completely converted into heat, but the reverse is not true: heat energy cannot be wholly transformed into work energy. Scientists and engineers have been able to exploit the principles of thermochemistry to develop technologies ranging from hot/cold packs to gasoline powered combustion engines.

For a closed system, the change in internal energy (ΔU) is related to heat (Q) and work (W) as follows:

$$\Delta U = Q + W \quad \Delta U = Q + W$$

This means that the total energy within a system is affected by the sum of two possible energy transfers: heat and work.

Bond Energy and Enthalpy

Bond Energy

Bond energy is the measure of bond strength. In order to turn one mole of a molecule into its constituent atoms, an amount of heat equal to the bond energy needs to be put into the system.

Learning Objectives

Identify the relationship between bond energy and strength of chemical bonds

Key Takeaways

Key Points

- Values listed in tables of bond energy and bond length are averages taken over a variety of compounds that contain a specific atom pair.
- A plot of the potential energy of a two-atom system and the distance between the atoms reveals a distance at which the energy is at its minimum. This distance is the bond length between the atoms.
- The higher the bond energy associated with a specific atom pair, the stronger the bond is said to be, and the smaller the distance between the two atoms.

Key Terms

- equilibrium bond length: The average distance between two atoms when they are bonded to each other.
- Morse curve: A plot showing the dependence of the energy associated with a system of two atoms on the distance between them (referred to as the 'internuclear distance').
- enthalpy: In thermodynamics, a measure of the heat content of a chemical or physical system, measured under conditions of constant pressure.

□ **Bond Energy:** A measure of a chemical bond's strength. It is experimentally determined by measuring the heat (or enthalpy) required to break a mole of molecules into their constituent individual atoms.

The Energy Associated with a Chemical Bond

Bond energy is a measure of a chemical bond's strength, meaning that it tells us how likely a pair of atoms is to remain bonded in the presence of energy perturbations. Alternatively, it can be thought of as a measure of the stability gained when two atoms bond to each other, as opposed to their free or unbound states.

Bond energy is determined by measuring the heat required to break one mole of molecules into their individual atoms, and it represents the average energy associated with breaking the individual bonds of a molecule. The higher the bond energy, the 'stronger' we say the bond is between the two atoms, and the distance between them (bond length) is smaller.

For instance, the HO-H bond in a water molecule requires 493 kJ/mol to break and generate the hydroxide ion (OH⁻). Breaking the O-H bond in the hydroxide ion requires an additional 424 kJ/mol. Therefore, the bond energy of the covalent O-H bonds in water is reported to be the average of the two values, or 458.9 kJ/mol. These energy values (493 and 424 kJ/mol) required to break successive O-H bonds in the water molecule are called 'bond dissociation energies,' and they are different from the bond energy. The bond energy is the average of the bond dissociation energies in a molecule.

The exact properties of a specific kind of bond are determined in part by the nature of the other bonds in the molecule; for example, the energy and length of the C-H bond will vary depending on what other atoms are bonded to the carbon atom. Similarly, the C-H bond length can vary by as much as 4% between different molecules. For this reason, the values listed in tables of bond energy and bond length are usually averages taken over a variety of compounds that contain a specific atom pair.

We can apply bond energy values to determine the enthalpy of a compound's formation, ΔH_f , which can be roughly approximated by simply adding tabulated values for the bond energies of all created bonds. The accuracy of this method is within a few percent of the experimentally determined ΔH_f values.

Energy Between Two Atoms as a Function of Internuclear Distance

A Morse curve shows how the energy of a two atom system changes as a function of internuclear distance.

Morse curve: Plot of potential energy vs distance between two atoms. The bond energy is energy that must be added from the minimum of the 'potential energy well' to the point of zero energy, which represents the two atoms being infinitely far apart, or, practically speaking, not bonded to each other.

At large distances, the energy is zero, meaning that there is no interaction. This agrees with our understanding that two atoms placed infinitely far apart do not interact with each other in any meaningful way, or at the very least we can say that they are not bonded to each other. At

internuclear distances in the order of an atomic diameter, attractive forces dominate. At very small distances between the two atoms, the force is repulsive and the energy of the two atom system is very high. The attractive and repulsive forces are balanced at the minimum point in the plot of a Morse curve.

The internuclear distance at which the energy minimum occurs defines the equilibrium bond length. This bond length represents an 'equilibrium' value because thermal motion causes the two atoms to vibrate about this distance, much like a spring vibrates back and forth around its unstretched, or equilibrium distance.

A Morse curve will have different energy minima and distance dependence for bonds formed between different pairs of atoms. In general, the stronger the bond between two atoms, the lower the energy minimum is and the smaller the bond length.

The bond energy is the amount of work that must be done to pull two atoms completely apart; in other words, it is the same as the depth of the "well" in the potential energy curve.

Bond Enthalpy

Bond enthalpy is defined as the enthalpy change when a covalent bond is cleaved by homolysis.

Learning Objectives

Describe the changes in enthalpy accompanying the breaking or formation of a bond

Key Takeaways

Key Points

- Breaking a bond requires the input of energy (positive change in enthalpy); energy is released (negative change in enthalpy) when forming a bond.
- Bond enthalpy, or dissociation energy, is defined as the standard enthalpy change when a bond is cleaved by homolysis, with reactants and products of the homolysis reaction at 0 K (absolute zero).

Key Terms

- enthalpy: A measure of the heat energy content in a thermodynamic system.
- homolysis: The symmetrical breaking of a chemical bond within a molecule so that each of the resulting fragments retains one of the originally bonded electrons.
- exothermic: A chemical reaction that releases heat energy to its surroundings.
- endothermic: A chemical reaction that absorbs heat energy from its surroundings.

Energy Transfer During Breaking or Formation of Bonds

Change in Enthalpy

Enthalpy is a measure of the total heat energy content in a thermodynamic system, and it is practically used to describe energy transfer during chemical or physical processes in which the pressure remains constant.

The total enthalpy, H , of a system cannot be measured directly. Thus, the change in enthalpy, ΔH , is a more useful quantity than its absolute value. The change (ΔH) is positive in endothermic reactions because the products of the reaction have a greater enthalpy than the

reactants, and heat is absorbed by the system from its surroundings. The change in enthalpy is negative in exothermic processes, because energy is released from the system into its surroundings.

Generally, a positive change in enthalpy is required to break a bond, while a negative change in enthalpy is accompanied by the formation of a bond. In other words, breaking a bond is an endothermic process, while the formation of bonds is exothermic.

Bond Enthalpy or Dissociation Energy

Bond enthalpy, also known as bond dissociation energy, is defined as the standard enthalpy change when a bond is cleaved by homolysis, with reactants and products of the homolysis reaction at 0 K (absolute zero).

Homolysis of a chemical bond: A two-electron covalent bond is equally split when bond breaking, with each resulting fragment having one electron from the original shared pair. Notice that the products are free-radicals.

For instance, the bond enthalpy, or bond-dissociation energy, for one of the C-H bonds in ethane (C₂H₆) is defined by the process:



$$\Delta H = 101.1 \text{ kcal/mol} (423.0 \text{ kJ/mol})$$

The strength of bonds between different atoms varies across the periodic table and is well documented.

Bond dissociation energy: Bond dissociation energies for different element pairings are listed. It is evident that bond strength varies significantly for different combinations of elements in the periodic table.

Each bond in a molecule has its own bond dissociation energy, so a molecule with four bonds will require more energy to break the bonds than a molecule with one bond. As each successive bond is broken, the bond dissociation energy required for the other bonds changes slightly.

Bond Lengths

Bond length between two atoms depends on factors such as the orbital hybridization and the electronic nature of the components.

Learning Objectives

Compare bond lengths using knowledge of periodic trends

Key Takeaways

Key Points

- Bond length is the experimentally determined average distance between two bonded atoms.
- Bonded atoms vibrate due to thermal energy available in the surroundings.
- Bond lengths are typically in the range of 100-200 pm (1-2 Å).
- As a general trend, bond length decreases across a row in the periodic table and increases down a group.
- Atoms with multiple bonds between them have shorter bond lengths than singly bonded

ones.

Key Terms

□ **bond length:** In molecular geometry, bond length or bond distance is the average distance between nuclei of two bonded atoms in a molecule.

□ **covalent radius:** The radius of an atom when covalently bonded to other atoms.

The Distance Between Two Bonded Atoms

The distance between two atoms participating in a bond, known as the bond length, can be determined experimentally. X-ray diffraction of molecular crystals allows for the determination of the three-dimensional structure of molecules and the precise measurement of internuclear distances. Various spectroscopic methods also exist for estimating the bond length between two atoms in a molecule.

Bonds Are Not Static Structures

The bond length is the average distance between the nuclei of two bonded atoms in a molecule. This is because a chemical bond is not a static structure, but the two atoms actually vibrate due to thermal energy available in the surroundings at any non-zero Kelvin temperature. A bond can be modeled as two balls connected by a spring: stretching or compressing the spring initiates a back-and-forth motion with respect to the equilibrium positions of the balls. Measured bond lengths are the distance between those unperturbed, or equilibrium, positions of the balls, or atoms.

Ball-and-spring model of a chemical bond: A bond between two atoms can be thought of as a spring with two balls attached to it. Any stretch or compression of the spring will initiate oscillations of the atoms with respect to their equilibrium (unperturbed) positions. The potential energy function for this system is also indicated. The minimum energy occurs at the equilibrium distance r_0 , which is where the bond length is measured.

Bond Length Values

Bond lengths have traditionally been expressed in Ångstrom units, but picometers are sometimes preferred ($1 \text{ Å} = 10^{-10} \text{ m} = 100 \text{ pm}$). Bonds lengths are typically in the range of 1-2 Å, or 100-200 pm. Even though the bond vibrates, equilibrium bond lengths can be determined experimentally to within $\pm 1 \text{ pm}$. The actual distance between two atoms in a molecule depends on factors such as the orbital hybridization and the electronic nature of its components.

Bonds involving hydrogen can be quite short; the shortest bond of all, H-H, is only 74 pm. The covalent radius of an atom is determined by halving the bond distance between two identical atoms.

Based on data for the H_2 molecule, the covalent radius of H is 37 pm. Covalent radii can be used to estimate the bond distance between two different atoms; it is the sum of the individual covalent radii.

Periodic Trends

Generally, when we consider a bond between a given atom and a varying atomic bonding partner, the bond length decreases across a period in the periodic table, and increases down a group. This trend is identical to that of the atomic radius.

Atoms with multiple bonds between them have shorter bond lengths than singly bonded ones; this is a major criterion for experimentally determining the multiplicity of a bond. For example, the bond length of C–CC–C is 154 pm; the bond length of C=CC=C is 133 pm; and finally, the bond length of C≡CC≡C is 120 pm.

Processing math: 47%

Standard Enthalpy of Formation and Reaction

Standard States and Standard Enthalpy Changes

The standard enthalpy of formation refers to the enthalpy change when one mole of a compound is formed from its elements.

Learning Objectives

Calculate the standard enthalpy of formation

Key Takeaways

Key Points

- The standard state of a material is a reference point for the material's thermodynamic state properties such as enthalpy, entropy, Gibbs free energy, etc. It is used to calculate the material's properties under different conditions and is denoted as H°_f .
- The standard state for a gas is the hypothetical state at 1 bar; for liquids and solids, the pure substance at 1 bar; for elements, the most stable allotrope of the element; and for a substance in solution (solute), concentration at 1 M and 1 bar.
- The standard enthalpy of formation is the change in enthalpy that accompanies the formation of one mole of the compound from its elements. The standard enthalpy of reaction occurs in a system when one mole of matter is transformed by a chemical reaction.

Key Terms

- standard state: In chemistry, a reference point used to calculate a material's (pure substance, mixture, or solution) properties under different conditions.
- standard enthalpy of formation: The change in enthalpy that accompanies the formation of one mole of a compound from its elements, with all substances in their standard states; also called "standard heat of formation."
- enthalpy of solution: The heat association with dissolving a particular solute in a particular solvent.

Standard States

In chemistry, the standard state of a material, be it a pure substance, mixture, or solution, is a reference point used to calculate its properties under different conditions. In principle, the choice of standard state is arbitrary, although the International Union of Pure and Applied Chemistry (IUPAC) recommends a conventional set of standard states for general use. A standard pressure of 1 bar (101.3 kilopascals) has been accepted.

Strictly speaking, temperature is not part of the definition of a standard state; the standard state of a gas is conventionally chosen to be 1 bar for an ideal gas, regardless of the temperature. However,

most tables of thermodynamic quantities are compiled at specific temperatures, most commonly 298.15 K (exactly 25°C) or, somewhat less commonly, 273.15 K (exactly 0°C).

Standard states for atomic elements are given in terms of the most stable allotrope for each element.

For example, white tin and graphite are the most stable allotropes of tin and carbon, respectively. Therefore, they are used as standard states or reference points for calculations of different thermodynamic properties of these elements.

Tin: White tin (on the left) is the most stable allotrope of tin, and is used as its standard state for thermodynamic calculations.

The standard state should not be confused with standard temperature and pressure (STP) for gases,

or with the standard solutions used in analytical chemistry. Standard states are often indicated in textbooks by a circle with a horizontal bar $\text{H}\ominus\text{f}$.

Graphite: Graphite is the most stable state of carbon and is used in thermochemistry to define the heat of formation of carbon compounds.

Standard Enthalpy of Formation

The standard enthalpy of formation, or standard heat of formation, of a compound is the change in enthalpy that accompanies the formation of one mole of the compound from its elements in their standard states. For example, the standard enthalpy of formation for carbon dioxide would be the change in enthalpy for the following reaction:



Note that standard enthalpies of formation are always given in units of kJ/mol of the compound formed.

Standard Enthalpy of Reaction

The standard enthalpy of reaction is the enthalpy change that occurs in a system when a chemical reaction transforms one mole of matter under standard conditions.

Learning Objectives

Demonstrate how to directly calculate the standard enthalpy of reaction

Key Takeaways

Key Points

- The standard enthalpy of reaction, $\Delta \text{H}\ominus\text{rxn}$, can be calculated by summing the standard enthalpies of formation of the reactants and subtracting the value from the sum of the standard enthalpies of formation of the products.
- The following equation can be used to calculate the standard enthalpy of reaction:

$$\Delta \text{H}\ominus\text{rxn} = \sum \Delta \text{H}\ominus\text{f}\{\text{products}\} - \sum \Delta \text{H}\ominus\text{f}\{\text{reactants}\}.$$
- The enthalpy of reaction is calculated under standard conditions (STP).

Key Terms

- standard enthalpy of reaction: The enthalpy change that occurs in a system when one mole

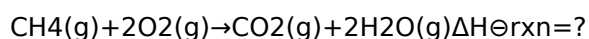
of matter is transformed by a chemical reaction under standard conditions.

The standard enthalpy of reaction, $\Delta H^\circ_{\text{rxn}}$, is the change in enthalpy for a given reaction calculated from the standard enthalpies of formation for all reactants and products. The change in enthalpy does not depend upon the particular pathway of a reaction, but only upon the overall energy level of the products and reactants; enthalpy is a state function, and as such, it is additive. In order to calculate the standard enthalpy of a reaction, we can sum up the standard enthalpies of formation of the reactants and subtract this from the sum of the standard enthalpies of formation of the products. Stated mathematically, this gives us:

$$\Delta H^\circ_{\text{rxn}} = \sum \Delta H^\circ_f \{\text{products}\} - \sum \Delta H^\circ_f \{\text{reactants}\}$$

Calculating the Standard Enthalpy of Reaction

Calculate the standard enthalpy of reaction for the combustion of methane:



In order to calculate the standard enthalpy of reaction, we need to look up the standard enthalpies of formation for each of the reactants and products involved in the reaction. These are typically found in an appendix or in various tables online. For this reaction, the data we need is:

$$\Delta H^\circ_f \{\text{CH}_4(\text{g})\} = -75 \text{ kJ/mol}$$

$$\Delta H^\circ_f \{\text{O}_2(\text{g})\} = 0 \text{ kJ/mol}$$

$$\Delta H^\circ_f \{\text{CO}_2(\text{g})\} = -394 \text{ kJ/mol}$$

$$\Delta H^\circ_f \{\text{H}_2\text{O}(\text{g})\} = -284 \text{ kJ/mol}$$

Note that because it exists in its standard state, the standard enthalpy of formation for oxygen gas is 0 kJ/mol. Next, we sum up our standard enthalpies of formation. Keep in mind that because the units are in kJ/mol, we need to multiply by the stoichiometric coefficients in the balanced reaction equation.

$$\sum \Delta H^\circ_f \{\text{products}\} = \Delta H^\circ_f \{\text{CO}_2(\text{g})\} + 2 \Delta H^\circ_f \{\text{H}_2\text{O}(\text{g})\} = (-394) + (2)(-284) = -962 \text{ kJ/mol}$$

$$\sum \Delta H^\circ_f \{\text{reactants}\} = \Delta H^\circ_f \{\text{CH}_4(\text{g})\} + 2 \Delta H^\circ_f \{\text{O}_2(\text{g})\} = (-75) + (2)(0) = -75 \text{ kJ/mol}$$

Now, we can find the standard enthalpy of reaction:

$$\Delta H^\circ_{\text{rxn}} = \sum \Delta H^\circ_f \{\text{products}\} - \sum \Delta H^\circ_f \{\text{reactants}\} = (-962) - (-75) = -887 \text{ kJ/mol}$$

As we would expect, the standard enthalpy for this combustion reaction is strongly exothermic.

A calculation of standard enthalpy of reaction ($\Delta H^\circ_{\text{rxn}}$) from standard heats of formation (ΔH°_f): A standard enthalpy of reaction ($\Delta H^\circ_{\text{rxn}}$) problem, involving ethylene and oxygen as reactants to yield carbon dioxide and gaseous water, is shown.

Hess's Law

Hess's Law sums the changes in enthalpy for a series of intermediate reaction steps to find the overall change in enthalpy for a reaction.

Learning Objectives

Employ Hess's law to determine $\Delta H^\circ_{\text{rxn}}$

Key Takeaways

Key Points

□ Hess's law states that the standard reaction enthalpy is the sum of the standard enthalpies of the intermediate reactions into which the overall reaction can be divided, while each occurs at the same temperature.

□ Enthalpy change for a reaction is independent of the number of ways a product can be obtained, if the initial and final conditions are the same.

□ Negative enthalpy change for a reaction indicates exothermic process, while positive enthalpy change corresponds to endothermic process.

Key Terms

□ Hess's law: States that, if an overall reaction takes place in several steps, its standard reaction enthalpy is the sum of the standard enthalpies of the intermediate reactions, at the same temperature.

□ law of conservation of energy: States that the total amount of energy in any isolated system remains constant; energy cannot be created or destroyed, although it can change forms.

Derivation of Hess's Law

Hess's law is a relationship in physical chemistry named after Germain Hess, a Swiss-born Russian chemist and physician. This law states that if a reaction takes place in several steps, then the standard reaction enthalpy for the overall reaction is equal to the sum of the standard enthalpies of the intermediate reaction steps, assuming each step takes place at the same temperature.

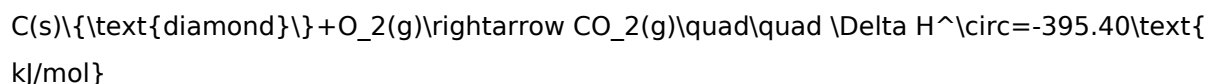
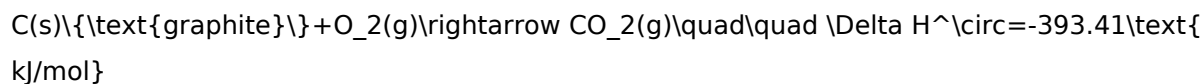
Hess's law derives directly from the law of conservation of energy, as well as its expression in the first law of thermodynamics. Since enthalpy is a state function, the change in enthalpy between products and reactants in a chemical system is independent of the pathway taken from the initial to the final state of the system. Hess's law can be used to determine the overall energy required for a chemical reaction, especially when the reaction can be divided into several intermediate steps that are individually easier to characterize. Negative enthalpy change for a reaction indicates exothermic process, while positive enthalpy change corresponds to endothermic process.

Graphical representation of Hess's law: The net reaction here is A being converted into D, and the change in enthalpy for that reaction is ΔH . However, we can see that the net reaction is a result of A being converted into B, which is then converted into C, which is finally converted into D. By Hess's law, the net change in enthalpy of the overall reaction is equal to the sum of the changes in enthalpy for each intermediate transformation: $\Delta H = \Delta H_1 + \Delta H_2 + \Delta H_3$.

Calculating Standard Enthalpies of Reaction Using Hess's Law

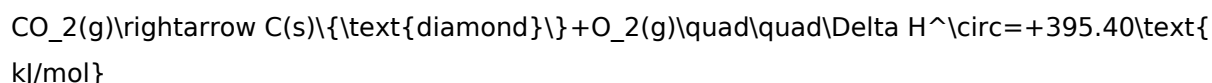
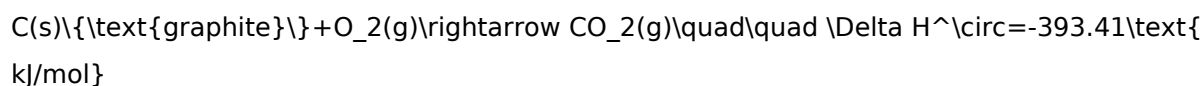
$\text{C(s)}_{\text{graphite}} \rightarrow \text{C(s)}_{\text{diamond}} \quad \Delta H_{\text{rxn}} = ?$

Turning graphite into diamond requires extremely high temperatures and pressures, and therefore is impractical in a laboratory setting. The change in enthalpy for this reaction cannot be determined experimentally. However, because we know the standard enthalpy change for the oxidation for these two substances, it is possible to calculate the enthalpy change for this reaction using Hess's law. Our intermediate steps are as follows:

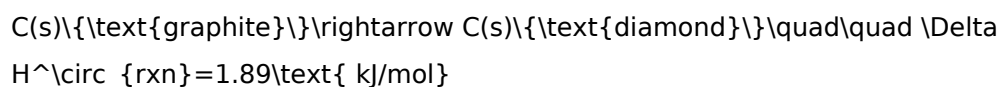


In order to get these intermediate reactions to add to our net overall reaction, we need to reverse the second step. Keep in mind that when reversing reactions using Hess's law, the sign of ΔH will change. Sometimes, you will need to multiply a given reaction intermediate through by an integer.

In such cases, you need always multiply your ΔH value by that same integer. Restating the first equation and flipping the second equation, we have:



Adding these equations together, carbon dioxides and oxygens cancel, leaving us only with our net equation. By Hess's law, we can sum the ΔH values for these intermediate reactions to get our final value, $\Delta H^\circ_{\text{rxn}}$.



Hess's law lesson: This lesson uses two methods to find the heat of reaction for a given reaction. First it looks at combining reactions according to Hess's law and their heats of reaction, and then it discusses using standard heats of formation of the reactants and products to find the overall heat of reaction.

Heat of Solution

Heat of solution refers to the change in enthalpy when a solute is dissolved into a solvent.

Learning Objectives

Define heat of solution

Key Points

- Enthalpy of solution, or heat of solution, is expressed in kJ/mol, and it is the amount of heat energy that is released or absorbed when a solution is formed.

- There are three steps in solvation: the breaking of bonds between solute molecules, the breaking of intermolecular attractions between solvent molecules, and the formation of new solute-solvent attractive bonds. Energy is absorbed during the first two steps, and it is released during the last step.

□ Depending on the relative amounts of energy required to break bonds initially, as well as how much is released upon solute-solvent bond formation, the overall heat of solution can either be endothermic or exothermic.

Key Terms

□ heat of solution: The enthalpy change associated with the dissolution of a substance in a solvent at constant pressure, resulting in infinite dilution.

□ solvation: The process of attraction and association of molecules of a solvent with molecules or ions of a solute; also called dissolution.

The heat of solution, also referred to the enthalpy of solution or enthalpy of dissolution, is the enthalpy change associated with the dissolution of a solute in a solvent at constant pressure, resulting in infinite dilution. The heat of solution, like all enthalpy changes, is expressed in kJ/mol for a reaction taking place at standard conditions (298.15 K and 1 bar).

Three-Step Process of Dissolution

The heat of solution can be regarded as the sum of the enthalpy changes of three intermediate steps:

1. The breaking of bonds within the solute, such as the electrostatic attraction between two ions (endothermic)
2. The breaking of intermolecular attractive forces within the solvent, such as hydrogen bonds (endothermic)
3. The formation of new attractive solute-solvent bonds in solution (exothermic)

The value of the overall heat of solution, $\Delta H^{\circ}_{\text{sol}}$, is the sum of these individual steps. Depending on the relative signs and magnitudes of each step, the overall heat of solution can be either positive or negative, and therefore either endothermic or exothermic. This depends entirely on if more energy was used to break the solute-solute and solvent-solvent bonds, or if more energy was released when solute-solvent bonds were formed.

If more energy is released in making bonds than is used in breaking bonds, the overall process is exothermic, and ΔH_{sol} is negative. If more energy is used in breaking bonds than is released upon solute-solvent bond formation, then the overall process is endothermic, and ΔH_{sol} is positive.

Examples

□ Dissolution of sodium chloride (table salt) in water is endothermic. This is because the amount of energy used to break apart the hydrogen bonding interactions between water molecules, as well as the energy used to break apart the electrostatic attractions between sodium and chloride ions, is greater than the amount of energy released when new solute-solvent attractions are formed between water molecules and aqueous ions in solution.

□ Dissolving potassium hydroxide is exothermic. This is because more energy is released upon formation of solute-solvent bonds than was required to break apart the hydrogen bonds in water, as well as the ionic bonds in KOH.

Dissolution of NaCl in water: Dissolution of sodium chloride in water is endothermic. Solute-solvent attractive bond formation (the exothermic step in the process of solvation) is indicated by

dashed lines.